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Denis Day



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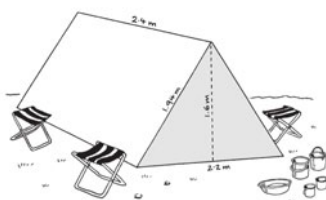
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MATHS in War & Conflict



Denis Day

This book was commissioned by Jennifer Gough, project management, editing and typesetting was by Alison Stone, illustrations by Eddy Crosby, and Gail McManus Graphics designed the cover.

Denis Day completed a mechanical engineering diploma and worked as a production engineer before completing his education qualifications (Master of Education). For more than 30 years he has been teaching maths and science in socially disadvantaged schools and he has also taught in private schools. Denis has presented numerous workshops at the Maths Association of Victoria's annual maths conference and regional conferences, and developed many curriculum materials.

His particular strength is in devising high interest activities which immediately provide a meaningful context for the mathematical concepts being studied – making learning both more effective and much more enjoyable for students.

Denis's interests include playing and coaching tennis (Level 2 tennis coach), saltwater fishing and Aussie Rules football (Level 2 umpire).

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Maths in War & Conflict is a collection of photocopiable worksheets about some of the most significant events in Australian history. Each topic has been written to interest and engage students in Years 9–10, and features content that requires students to apply key maths skills such as number, percentages, reading maps and drawing and interpreting graphs. The activities will help students to see how the skills of maths can be used to understand real world events.

Many of the activities in this book utilise data from a variety of sources. This data may have been reconfigured or altered for the purpose of understanding an essential mathematical concept. Whilst approximately correct, this data is not intended to be accurate and students should not use any data or statistics as source material for other projects.

The units of work are presented in chronological order. However, each unit stands alone and they do not need to be completed sequentially. Unless advised otherwise, students are asked to round their answers to two decimal places. For some topics an atlas and calculator would be useful. Answers are provided to make marking quick and easy.

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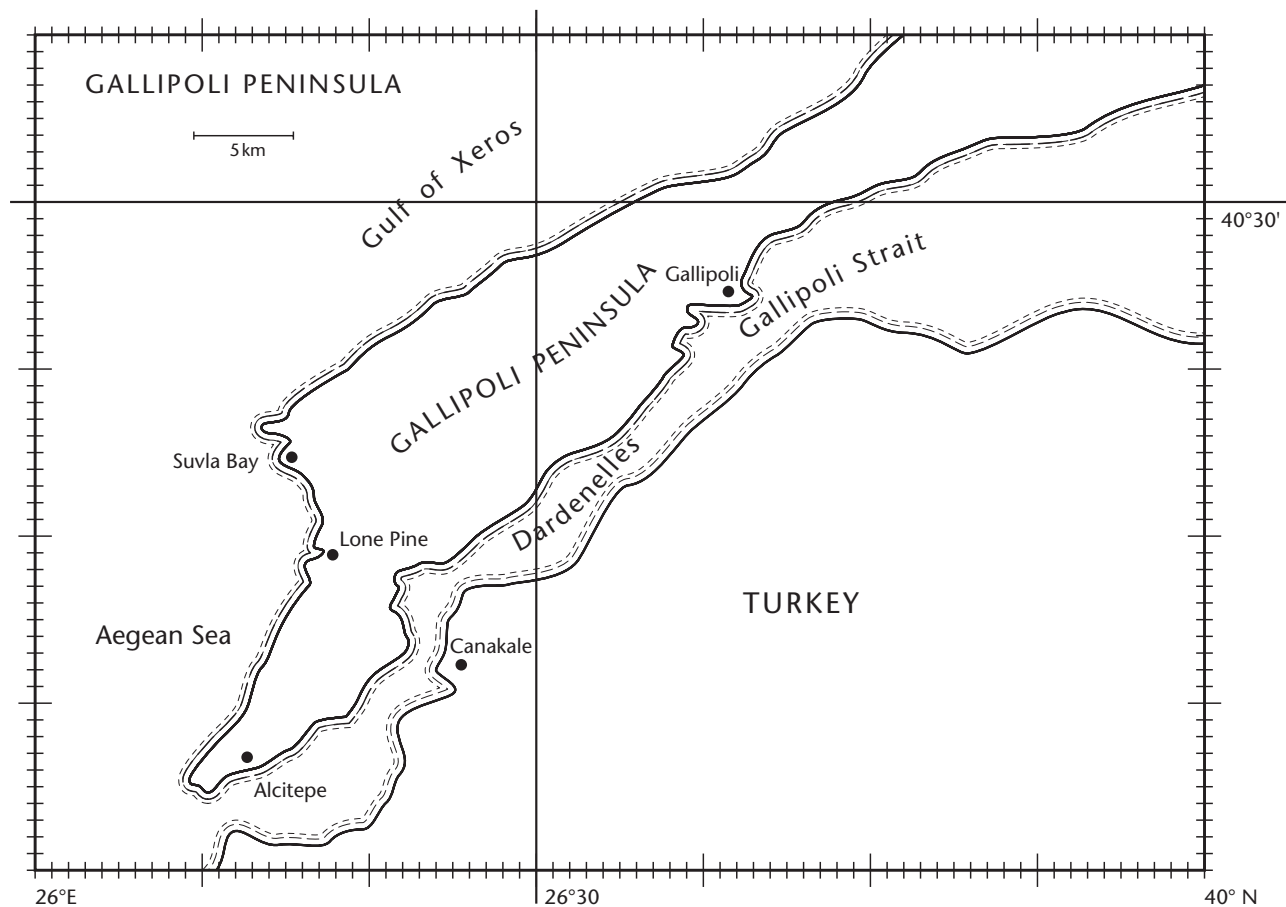
Gallipoli and the Anzacs



1

Since 1915, one day in the year has involved the whole of Australia in solemn ceremonies of remembrance, gratitude and national pride. That day is Anzac Day – 25 April.

Ten weeks after the beginning of World War 1, Australia and New Zealand sent 30 000 troops to Egypt for combat training. It was here that the Australian and New Zealand Army Corps acronym ANZAC was first used. At 3 am, on April 25 1914 just off the shores of Gallipoli in Turkey, in an area later to be known as Anzac Cove, three battleships gathered full of Anzac soldiers about to head into history. HMAS Queen, HMAS London and HMAS Prince of Wales released 48 boats or 'tows' at 3.30 am and the landings continued from 5.30 to 7.30 am. But something went wrong. The landings were over 2 km north of the intended area – a 6 km stretch of beach from about 1.6 km north of Gaba Tepe to a point near Fisherman's Hut.



- 1 Colour these areas on the map.

a Turkey	b Gallipoli Peninsula	c Gallipoli Strait
d Dardenelles	e Aegean Sea	f Gulf of Xeros
- 2 Use the scale on the map to work out:
 - a the length of the Dardenelles waterway from Alcitepe to Gallipoli
 - b the distance from Lone Pine to Suvla Bay.
- 3 Write the grid reference of:

a Suvla Bay	b Lone Pine	c Gallipoli	d Canakale	e Alcitepe.
-------------	-------------	-------------	------------	-------------

Gallipoli and the Anzacs (cont'd)



1

4 Label these locations on the map.

a Cape Helles 40°2'N 26°11'E

c Hell Spit 40°13'N 26°18'E

e Fishermans Hut 40°16'N 26°18'E

g Eceabat 40°11'N 26°21'E

b Gaba Tepe, 40°12'N 26°17'E

d Ari Burnu 40°14'N 26°18'E

f Kilitbari 40°9'N 26°22'E

h Sedulbahir 40°3'N 26°12'E

5 Mark these places on the map.

a Anzac Cove

b the beach the Anzacs landed on

c the position (and name) of each battleship

6 There are 15 000 men in a division and 5000 men in a brigade. A division can consist of several brigades. A corp consists of several divisions. How many men are in:

a a division of 3 brigades?

b a corp of 3 divisions?

c 5 brigades?

d 6 divisions?

7 24-hour time is commonly used in the army. Change these times to 24-hour (or army) time.

a 3.30 am

b 3.00 am

c 5.30 am

d 7.30 am

e 8.30 pm

8 Draw a timeline to show the events of Anzac Day.

9 How many boats were despatched from the battleships in the first wave?

10 After their first day, the men had suffered 2300 casualties and had covered only 900 m. Until they were evacuated, little changed from the ground captured on the first day of the landing. A space less than 2 km² in size became home to over 20 000 men. How much space did each soldier have in metres squared? Compare this to your classroom.

11 The table at right shows the approximate numbers of soldiers killed and wounded at Gallipoli.

	Nation	Died	Wounded	Total
a	United Kingdom	21 000	52 000	
b	France	10 000	17 000	
c	Australia	9 000	19 000	
d	New Zealand	3 000	5 000	
e	British India	1 400	3 400	
f	Newfoundland	50	90	
g	Turkey	87 000	165 000	
h	Total			

Complete the table.

12 How many soldiers were killed or wounded per day in the battle from 25 April to 20 December?



Besides the constant fear of death on the hills and dunes of Gallipoli, some logistical challenges included feeding the soldiers. Food supplies came from as far away as Greece and Egypt and perishables just wouldn't survive. The soldiers' food came in tins with the staples being corned beef (Bully Beef), rice, tea, cocoa, jam and 'hard tack'. Hard tack is basically a very hard biscuit made of flour, water, salt and sugar. It lasts longer than bread, although it was so hard soldiers would sometimes break their teeth trying to eat it.

Unlike bread, hard tack biscuits are very, very hard. Some soldiers preferred to grind them up with old ammunition shells, mix them with water and eat it as porridge.

The recipe below has been based on a recipe used during World War 1.

Ingredients (makes six biscuits)

- 190 g white flour
- 375 g wholemeal flour
- 40 g sugar
- 21 g milk powder
- good pinch salt
- 220 ml water



- 1 During World War 1 the metric unit of measurement was not in use. The British system of measurement was used. Rewrite the list of ingredients above, as it would have appeared in 1914, if 1 cup flour is 125 g, 1 tbsp sugar is 8 g, 1 tbsp milk powder is 7g and 1 cup water is 220 ml.
- 2 How many grams of milk powder in:
 - a 3 tablespoons?
 - b 6 tablespoons?
 - c 8 tablespoons?
 - d 13 tablespoons?



- 3 How many grams of white flour in:
 - a 2 cups?
 - b 3 cups?
 - c 4.5 cups?
 - d 6.2 cups?
- 4 How many cups of wholemeal flour in:
 - a 400 g?
 - b 800 g?
 - c 450 g?
 - d 620 g?
- 5 How many grams of sugar in:
 - a 1 tablespoon?
 - b 2 tablespoons?
 - c 6 tablespoons?
 - d $3\frac{1}{2}$ tablespoons?
- 6 The daily rations for a soldier was theoretically 3 ounces of cheese, $\frac{5}{8}$ ounce of tea, 4 ounces of jam, $\frac{1}{2}$ ounce of salt, $\frac{1}{36}$ ounce of pepper, four hard tack biscuits 3 ounces each, a 14 ounce can of bully beef and a quart (approximately 2.5 L) of water. Quite often, the soldiers were not provided with their daily ration of food.
 - a Change these rations into metric units of measurement, if 1 ounce is 28.35 g.
 - b Why do you think the soldiers were often not provided with their daily rations?
- 7 Calculate, by weight, the percentage of the soldiers' solid rations that was:
 - a jam
 - b tea
 - c hard tack
 - d bully beef.
- 8 There were more than 20 000 Anzac soldiers stationed at Gallipoli. Calculate the daily quantities of hard tack and jam that were needed to supply the soldiers. Write your answers in tonnes to one decimal place.
- 9 Convert measurements for hard tack biscuits to the metric system and draw a full size picture of a hard tack biscuit. The hard tack biscuits were in the shape of a square, each side measuring $3\frac{9}{16}$ inches and with a thickness of $\frac{7}{16}$ inches. Each biscuit had a pattern of 25 holes. (Remember that one inch is equal to 2.54 cm.)



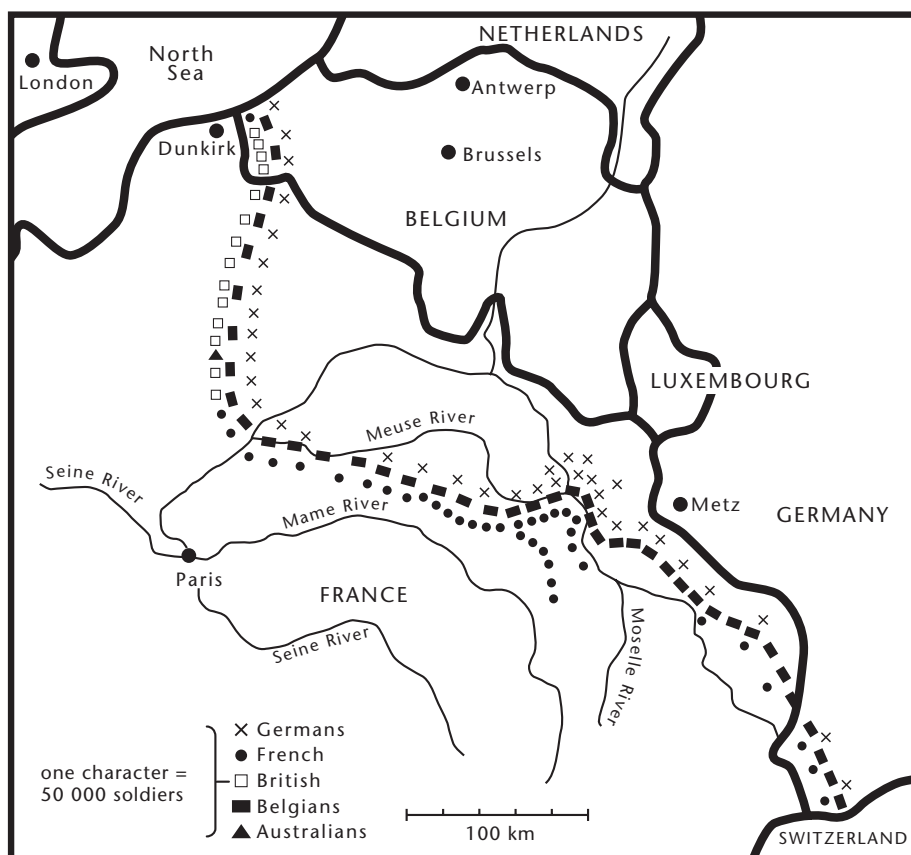


In World War 1 a series of trenches ran for hundreds of kilometres from the Belgian coast to the Swiss border. This became known as The Western Front. German and Allied soldiers dug the trenches as protection from the endless machine gun fire and bombing. At times the enemy trenches were just metres apart. Besides the constant danger of death from the shelling, the soldiers had to survive the challenge of winter as the trenches turned to mud.



- 1 Colour each country on the map.
- 2 How long, in kilometres and metres, were the trenches along the Western Front?
- 3 How many soldiers from each army were in battle on the Western Front?
- 4 Calculate the total number of soldiers involved in battle along the Western Front.
- 5 Calculate the percentage of soldiers from each army involved in the battle.
- 6 To the nearest kilometre, how far from Paris was the:
 - a Australian Army?
 - b German Army?
- 7 How far from Paris is:
 - a Dunkirk?
 - b London?
 - c Metz?
 - d Brussels?

- 8 Locate and colour these items on the map on page 9.
 - a forests, colour green
 - b lakes, colour blue
 - c marshes, colour brown
 - d railways, colour red
 - e rivers, colour blue
 - f canals, colour yellow



The Western Front (cont'd)



3

On the Western Front the Anzacs participated in major battles at the places listed below.

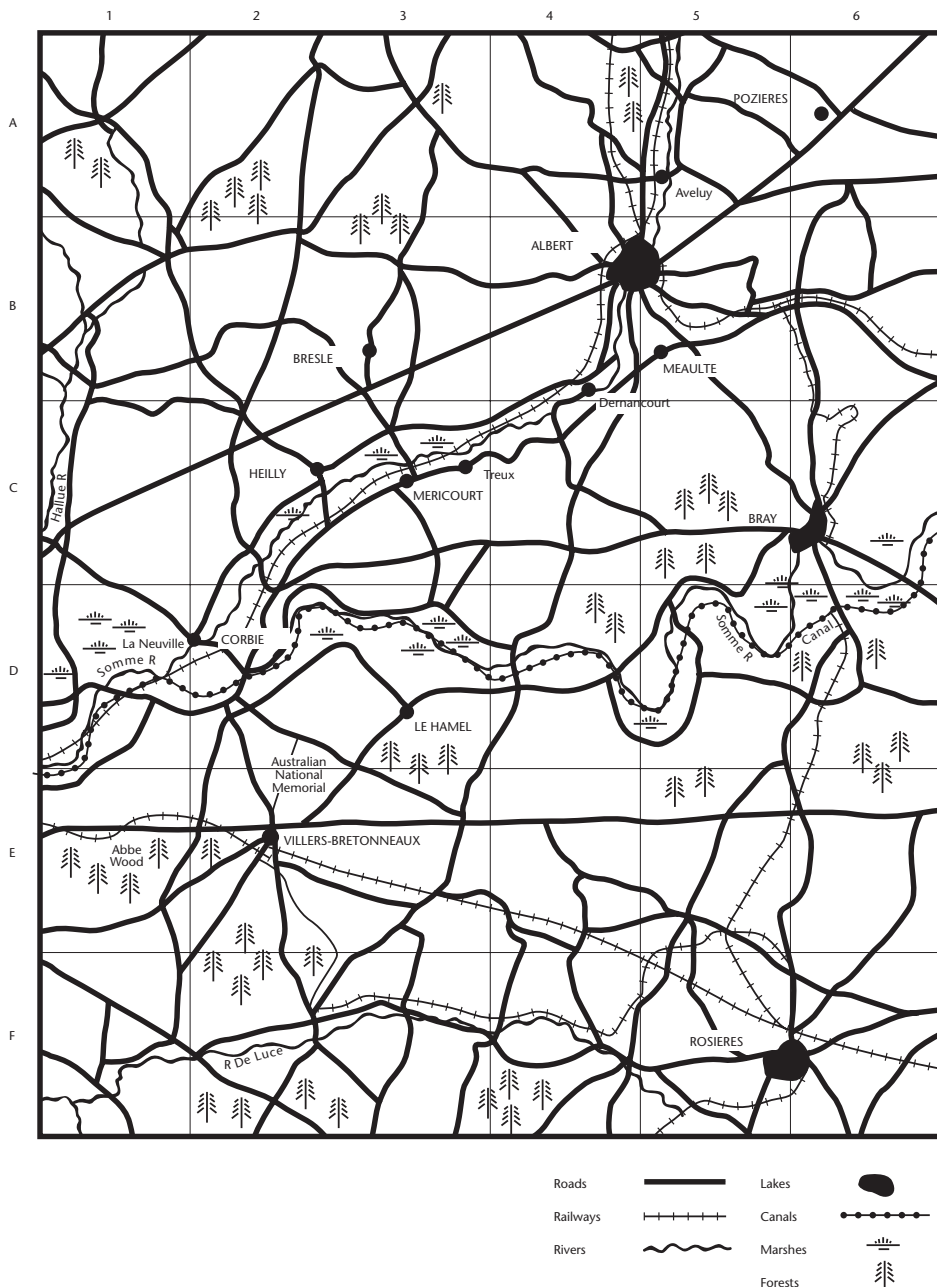
9 Locate these places on the map and write down their grid references.

a Dernancourt

b Le Hamel

c Pozières

d Villers-Bretonneaux



10 The phrase 'no man's land' refers to the area of ground between opposing army trenches. This area ranged from a kilometre to less than twenty metres in some places. Explain why the area between the trenches was called no man's land.

11 When the Anzacs arrived at the Western Front, the first thing they saw were the lines of wounded soldiers being taken to the rear. As they got closer, they could feel the earth shake, and hear the constant 'crump crump' of artillery shells. The sound was loud enough to make their ears ring, and became their companion for the next three years. Why do you think people still use the phrase 'all quiet on the Western Front'?



In 1898 one Lady Superintendent and 24 nurses formed the Army Nursing Service. This later became the Royal Australian Army Nursing Corps (RAANC). The RAANC were formed to care for Australian soldiers in times of war and its aftermath. The first group of nurses left Australia in September 1914 and over 2000 nurses served overseas during the war.

In September 1914, during World War 1, the first nurses left Australia. The nurses went wherever Australian and British troops served. Approximately 2140 served overseas, 420 served in Australia, 25 died and 390 were decorated.

- 1 Calculate the total number of Australian nurses who served.
- 2 Calculate the percentage of nurses who:

a served overseas

b served in Australia

c were decorated

d died.

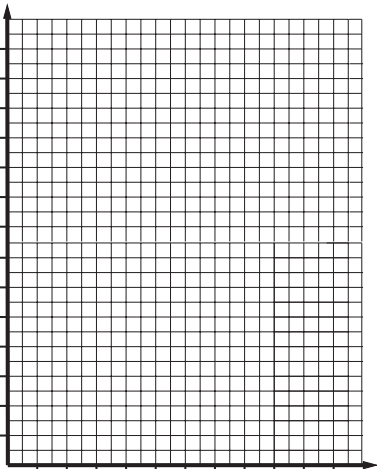
In World War 1 the RAANC sent Australian nurses to a ‘tent’ hospital in Cairo. Approximately 46 nurses had to manage a 1000-bed hospital set up entirely under tent canvas. Compare this to a modern metropolitan hospital in Australia with an average 700 beds and a staff of 670 nurses.

- 3 Complete the table.

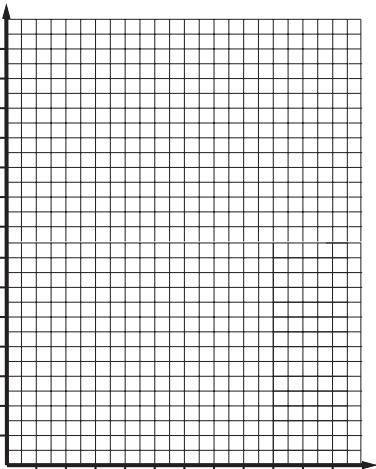
	Data	World War 1 hospital	Modern hospital
a	Number of beds		
b	Number of patients		
c	Number of nurses		
d	Patients per nurse		

- 4 Draw column graphs on the following grids for the data in the table to compare the two types of hospital.

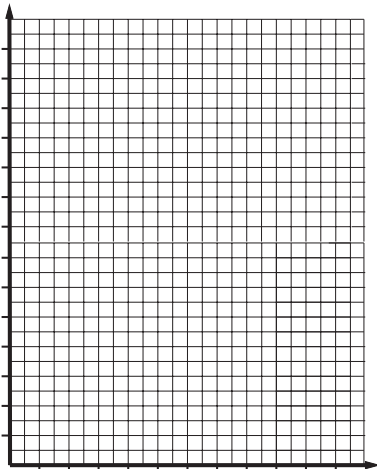
Patients



Nurses



Patients per nurse



Nurses were expected to be single or widowed. Some married women got through the recruiting checks, and some married during their period of service.



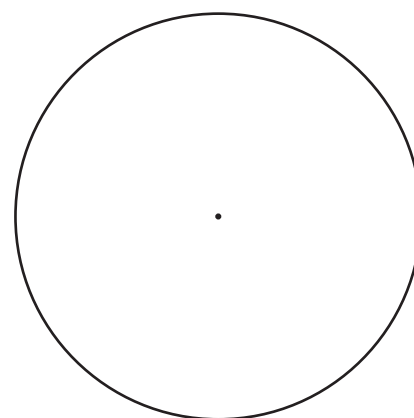
- 5 Complete the table and draw a sector graph using the template below.

	Age group (years)	Number of nurses	Fraction of each age group	Sector angle for each age group
a	Under 21	7		
b	21–30	1180		
c	31–40	950		
d	Older than 41	90		
e	Total			

- 6 Find the total number of nurses.

- 7 What is the percentage of nurses aged:

- a 21–30 years? b 31–40 years?
c younger than 21?



The nurses were treating soldiers who had been seriously wounded. Some soldiers had 12 operations or more. One hospital contained about 300 soldiers whose legs or arms had been amputated. Some nurses had to dress 30 leg stumps every morning and make up to 40 beds. Their lives were very stressful and some suffered nightmares and depression.

- 8 Calculate the percentage of the soldiers with stumps that each nurse has to dress each morning.

- a nurse Anne, 30 soldiers
b nurse Eileen, 18 soldiers
c nurse Monica, 25 soldiers
d nurse Bridget, 35 soldiers



- 9 If the hospital had 900 beds, what percentage of beds did these nurses have to make?

- a nurse Anne, 35 beds b nurse Mary, 45 beds
c nurse Hilary, 50 beds d nurse Margaret, 30 beds



Australian Red Cross was formed on 13 August 1914 exactly nine days after the outbreak of World War I. Lady Munro-Ferguson, who had worked for British Red Cross in Scotland and was the wife of the Governor-General, called on the wives of each State Governor to form a local committee in each capital. The women readily agreed and so Australian Red Cross was born.

During World War 2, Red Cross transported 40 000 pairs of socks, 25 000 shirts, 20 000 underpants, 10 000 pairs of shoes and boots, 10 000 pairs of pyjamas, 5000 overcoats and many other items people thought would be useful to soldiers on the front in Europe. The effort from many small communities ‘beggared the imagination’. By the end of the war Red Cross unit of Rutherglen had forwarded, among many other items, more than 1200 shirts, 8500 pairs of underpants, 1200 pairs of pyjamas, 2400 pairs of socks, 75 sheets, 770 pillow-cases, 450 bed sheets and 390 handkerchiefs.

1 Complete the table.

	Item	Number of items from Rutherglen	Number of items to Europe	Fraction of items from Rutherglen	Percentage of items from Rutherglen
a	Shirts				
b	Underpants				
c	Pyjamas				
d	Socks				

The volume of goods produced by Red Cross in five years, from 1914 to 1918, the funds raised and the personal efforts of thousands of people were quite unbelievable. A thirteen year old girl knitted 87 pairs of socks in a year, and a granny notched up 29 000 miles walking to collect funds. Australian Red Cross volunteers despatched nearly 400 000 food parcels and more than 36 000 clothing parcels.

Round answers to the nearest unit.

- 2

On average how many pairs of socks did the girl knit in a:

a

month?

b

week?
- 3

On average how many food parcels were despatched in a:

a

year?

b

month?
- 4

How many clothing parcels were sent to soldiers in a:

a

year?

b

week?
- 5

How far did the granny walk in kilometres if one mile is equal to 1.61 km?
- 6

On average how far did the granny walk (to the nearest km) in a:

a

year?

b

month?



Between 1941 and 1944, a Red Cross member made and bottled 8000 bottles of jam.



- 7** On average, how many bottles of jam did the member make in a:
- a** month? **b** year?

- 8** If a bottle of jam weighs 500 g, find the weight in kilograms of:
- a** 8000 bottles. **b** 600 bottles.

In 1915, Red Cross Transport Service commenced, driving soldiers who had returned on the hospital ships to their homes. By the end of 1916 the total number of cars was 2500 with more than half overseas, providing transport on the battlefields of France, Italy and East Africa.

- 9** Approximately how many cars did Red Cross have overseas?

During the First World War Australian Red Cross had a field force that served alongside the medical services of the armed forces. In total 347 men and 193 women served in 23 countries, as well as on hospital ships and aircraft carriers.

- 10** In total, how many people served for Red Cross Field Force?

- 11** What fraction were: **a** men? **b** women?

- 12** What percentage were: **a** women? **b** men?

A Red Cross ship, the *Centaur*, was torpedoed by an enemy submarine. On board were 74 crew, 65 army medical personnel, 192 soldiers from the 2/12 field ambulance and a pilot. Only 64 people survived.

- 13** How many people were on board the *Centaur*?

- 14** How many people died in the sinking of the *Centaur*?

- 15** What fraction were killed?

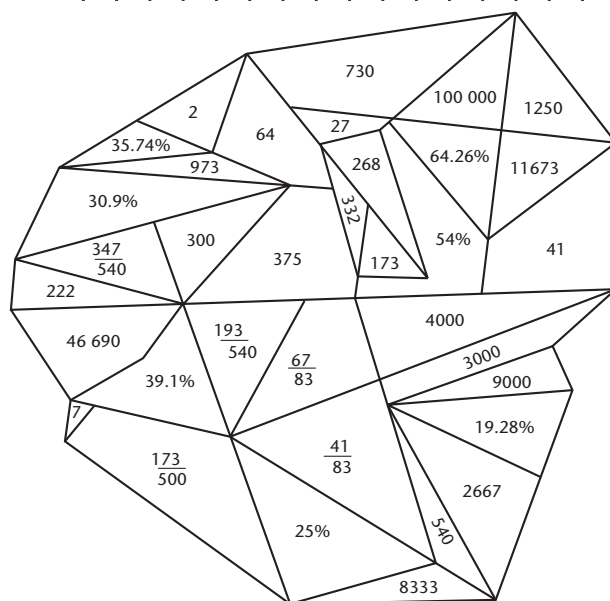
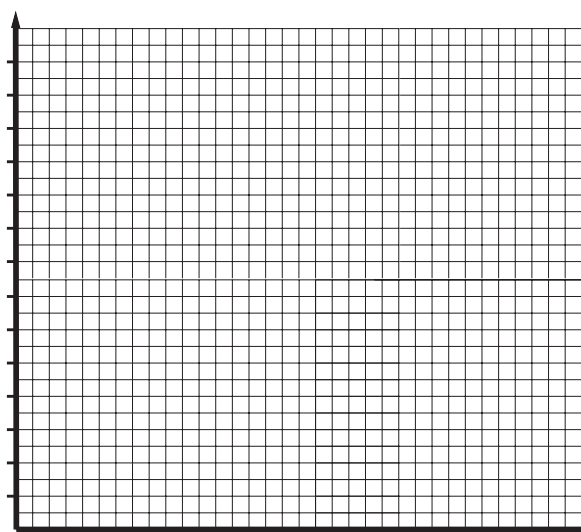
- 16** What percentage survived?

- 17** Draw a column graph to show the types of people that made up the crew.

The red cross is a protective emblem. It signifies 'don't shoot' and is essential for the survival of injured and sick members of all the military.

- 18** Find your answers to Q2–16 on the diagram. Colour the section red. Cut out each piece and use them to form the emblem of Red Cross, and if you have time, a pentagon.

- 19** List some things Red Cross do in the community today.



The Battle of the Coral Sea



6

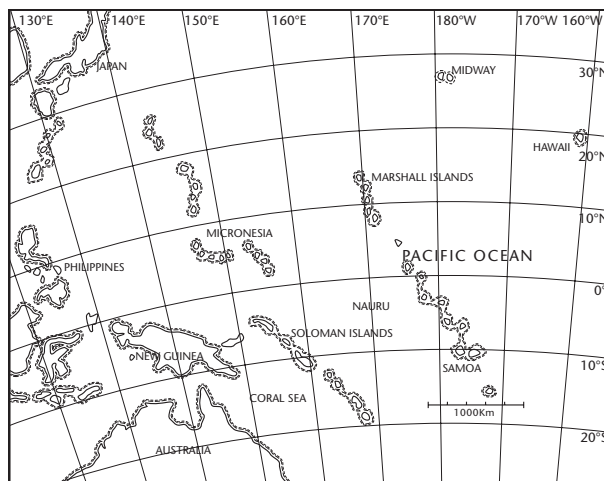
The Battle of the Coral Sea was actually a series of battles that pitted the forces of the United States and Australia against the Japanese on 4–8 May, 1942. This significant naval battle halted the seaborne invasion of Port Moresby by the Japanese. The ships did not engage in direct battle. It was a naval battle fought entirely by aircraft using the vessels as runways.

- 1 Mark these cities on the map and colour the land.

- a Townsville 19°S, 146°E
- b Tokyo 40°N, 140°E
- c Port Moresby 9°S, 147°E
- d Pearl Harbour 21°N, 158°W

- 2 Find these locations, provide a latitude and longitude and colour them.

- a The Philippines
- b Hawaii
- c Japan
- d Pacific Ocean
- e Coral Sea
- f Solomon Islands



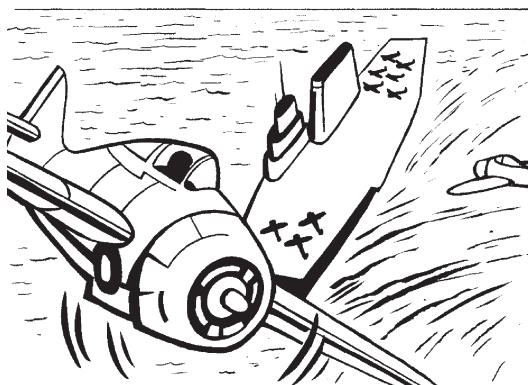
	Location	Distance	Travel time
a	Tokyo to Midway		
b	Hawaii to the Coral Sea		
c	Tokyo to Hawaii		
d	Townsville to Port Moresby		
e	Townsville to Tokyo		

The ships could travel at approximately 65 km/h.

- 3 Use the map to find the distance between locations and then calculate the time it would take a carrier to travel to that location. Record your results in the table.

During the Battle of the Coral Sea, two American carriers sent 54 dive bombers, 21 torpedo bombers and 24 fighters to strike the Japanese fleet. An American carrier could carry 95 aircraft. Forty-three planes were lost.

- 4 How many planes were sent to strike the Japanese fleet?
- 5 Of all the American planes, what percentage were sent to the battle?
- 6 Of the American planes that went to battle, what percentage survived?
- 7 Fighters make up 25% of a carrier's aircraft. How many fighters on average are on a carrier?
- 8 Midway, Pearl Harbour and the Coral Sea were zones of major conflict. Colour the areas of ocean around these zones.



The Battle of the Coral Sea (cont'd)



6

At Midway, three American carriers sank four Japanese carriers at a cost of one American carrier. In the whole battle, 126 out of 233 American planes and none of the 272 Japanese planes survived.

- 9 How many American planes were destroyed in the battle at Midway?
- 10 Of the American planes that participated in the battle, what percentage survived?

An aircraft carrier is a warship designed to allow the takeoff and landing of planes. The carrier has a flat upper deck that acts as a runway and a deck beneath the flight deck for storing and servicing the aircraft. The use of the aircraft carrier emerged just prior to World War 2 and was a necessity in naval operations providing air support to many military actions.

Super aircraft carriers stand approximately 75 m tall and stretch about 330 m from bow to stern. The MCG football ground is approximately 170 m long and 150 m wide.

- 11 Select a suitable scale and draw a column graph on the axes provided, to compare the size of the super aircraft carrier with the MCG football ground.

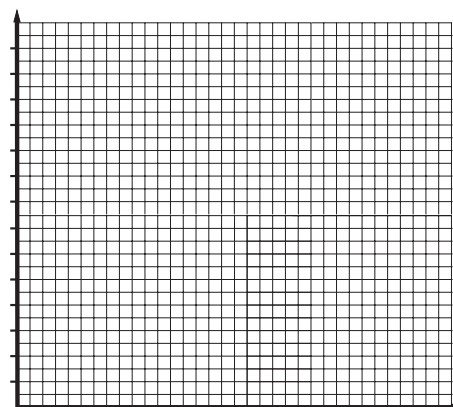
The super carriers can launch or land a plane every 25 seconds.

- 12 Calculate the average number of planes that can be launched or landed in:

- a 2 minutes
- b 10 minutes
- c 125 seconds.

- 13 Evaluate in minutes and seconds how long it would take to launch:

- a 5 planes
- b 6 planes
- c 10 planes
- d 3 planes.



Engineers estimate the following supplies are required to build super aircraft carriers.

- 14 Complete the table.

	Statistic	For 1 carrier	For 10 carriers
a	Weight of steel used		610 000 t
b	Weight of each anchor	31 t	
c	Weight of each link in anchor chain		1600 kg
d	Weight of each propeller	30 000 kg each at 4 per carrier	
e	Aviation fuel capacity	12.5 million L	
f	Number of crew		60 000
g	Number of aircraft	80	
h	Fresh water distilled per day		1 800 000 L
i	Cost	\$4.5 billion	

The Kokoda Trail



7

In World War 2, the Japanese army had advanced through Asia en route to Australia. Port Moresby, the capital of Papua New Guinea, was a significant base for both armies. Had the Japanese set up base in Port Moresby the Australian mainland would have been under severe threat of invasion. The Kokoda track was a seldom-used trail from the small coastal village of Buna on the north coast of Papua over the Owen Stanley Ranges to Port Moresby, passing through the village of Kokoda. In 1942 the Japanese tried to capture Port Moresby by sea attack and then overland through Kokoda. The seven-month resistance of this ultimately unsuccessful campaign is regarded by some as one of the most heroic defensive actions in military history by an Australian force.

1 Write down the latitude and longitude of these cities.

- a Tokyo b Port Moresby c Bali
d Darwin e Cairns f Manila

2 Name the country at each location.

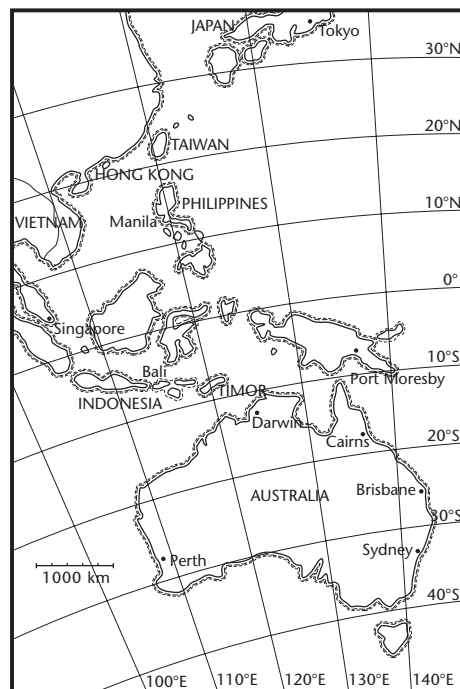
- a 25°S, 130°E b 10°N, 120°E c 10°S, 120°E
d 35°N, 135°E e 15°N, 110°E f 22°N, 122°E

3 Find the distance from Brisbane to:

- a Tokyo b Darwin c Port Moresby
d Cairns e Manila f Bali.

4 State the true bearing from Tokyo to:

- a Port Moresby b Manila
c Singapore d Brisbane.



5 The year is 2011. How many years has it been since the Kokoda Trail battle?

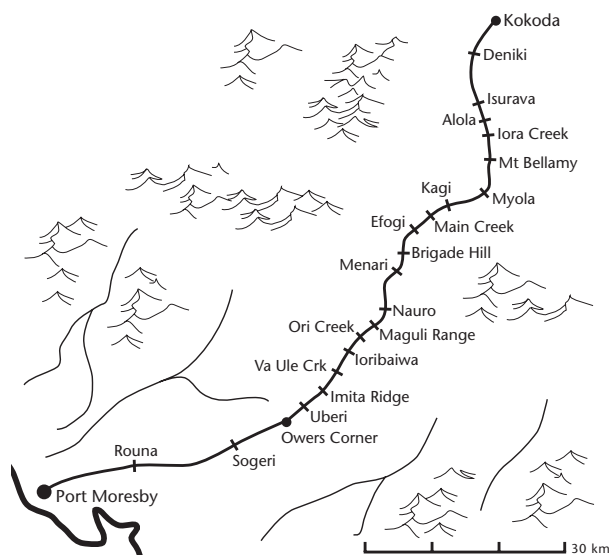
6 How many months is this?

7 Briefly explain why Kokoda was important to the Japanese and the Australian armies.

Height above sea level (in metres) for localities on the Kokoda Trail

Locality	Height
Owers Cnr	640
Uberi	440
Imita Ridge	860
Va Ule Crk	300
Ioribaiwa	610
Ori Crk	800
Maguli Rng	1350
Nauro	950
Menari	850
Brigade Hill	1470

Locality	Height
Efogi	1220
Main Crk	950
Kagi	1600
Myola	1900
Mt Bellamy	2200
Iora Crk	1420
Alola	1340
Isurava	1325
Deniki	366
Kokoda	340



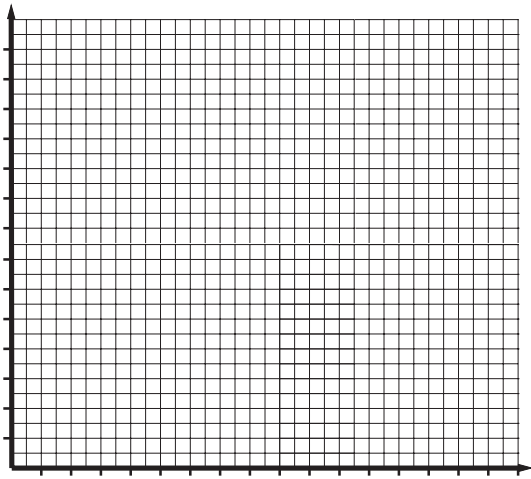
The Kokoda Trail (cont'd)



- 8 Colour the map.
- 9 The Kokoda Trail covers a distance of approximately 95 km. Use the scale on the map and tables over the page to calculate the distance between each locality, and the difference in height above sea level. Complete the table.

	Localities	Distance	Difference in sea level
a	Uberi to Imita Ridge		$860 - 440 = 420 \text{ m}$
b	Imita Ridge to Va Ule Creek		
c	Ori Creek to Maguli Range		
d	Menari to Brigade Hill		
e	Efogi to Main Creek		
f	Main Creek to Kagi		
g	Kagi to Myola		
h	Myola to Mt Belami		
i	Iora Creek to Isurava		
j	Isurava to Deniki		

- 10 Use your answers to Question 9 to plot points on a scatter diagram to represent the average slope of the terrain between each locality. Use the grid below.
- 11 Which section was the steepest?
- 12 Which section had the smallest slope?
- 13 The maximum slope for a wheel chair ramp is 1:7. Draw a line on the grid to represent this slope.
- 14 Draw a diagram to show a cross section of the Kokoda trail, from Port Moresby to Kokoda, showing features such as mountains etc.



The Rats of Tobruk



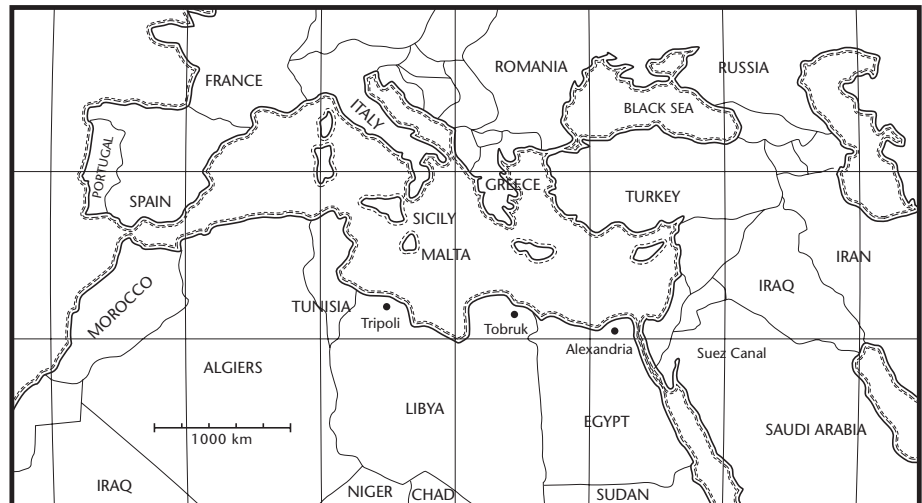
8

During World War 2, North Africa became a place of strategic importance to the Allied and German forces due to its position as a supply route. The Allied forces in North Africa included troops from Britain, Australia, New Zealand, United States and India. German troops were joined in their fight against the Allied forces by the Italian army.

- September 1940: Germans invade Egypt
- January 1941: Australians capture Tobruk and its port
- March 1941: Germans defeat Australians at El Agheila in Libya
- April 1941: Germans capture Libya (except for Tobruk)
- April 1941: Germans attack Tobruk
- June 1942: Germans threaten Egypt
- May 1943: The Allied forces evict Germans from Africa

- 1 Draw a timeline to represent these events.
- 2 Colour the countries and the seas on the map.

Germans and Italians reported 620 000 casualties in the campaign, the British and Australians lost 220 000 men. United States casualties in Tunisia totalled more than 18 500.



- 3 Calculate the total number of these reported casualties in the North African battles.
- 4 Calculate the fraction of the casualties from:
 - a Britain and Australia
 - b United States
 - c Germany and Italy.
- 5 Convert the fractions in Q4 to percentages.

The British and Australians won a decisive victory over the German and Italian forces at El Alamein in Egypt. The Germans and Italians had a total of 110 000 men and 500 tanks. The Allies had more than 200 000 men and more than 1000 tanks. This battle cost the Allied forces nearly 13 500 men; Germans and Italians lost 25 000 soldiers. The battle lasted for 13 days.

- 6 How many days (inclusive) did the battle last?
- 7 Calculate the total number of soldiers killed.
- 8 On average, how many soldiers were killed per day in the:
 - a Allied forces?
 - b German and Italian Army?
- 9 Calculate the percentage of soldiers killed from the:
 - a Allied forces
 - b German and Italian Army.

The Rats of Tobruk (cont'd)



8

10 What fraction of the tanks belonged to the:

- a** German and Italian Army? **b** Allied forces?

'Operation Supercharge' was an attack by British and New Zealand troops against the forces of Rommel – the leader of the German and Italian Army in North Africa. 124 tanks attacked the German lines but 75% of them were lost in a sandstorm.

11 How many tanks were destroyed by the German Army? How many tanks were left?

The Allied forces began damaging the supply lines to Rommel so that only 33% of what Rommel needed was getting through to him. During June he received 5000 tonnes of supplies compared with 34 000 in May and 400 vehicles compared with 2000 in May.

12 List some of the supplies you think Rommel may have needed.

13 Find the quantities of supplies and vehicles that Rommel actually needed.

14 Write the quantities Rommel was receiving in June as a percentage of the May quantities.

The Australian troops made up more than half of the Allied presence in Tobruk with a total strength of over 14 000 men. The remainder of the Allied force was made up of 12 000 British and Indian troops.

15 Calculate the total number of Allied troops at Tobruk.

16 What fraction of the troops were:

- a** Australians? **b** British and Indian?

17 What percentage of the troops were Australian?

The Australians gave themselves the nickname 'the Rats of Tobruk' after Radio Berlin described the Australians as 'caught like rats in a trap'. The Australians managed to hold off the German forces for 250 days.

18 For how many months and days did the Australians hold off the Germans (assume 30 days in a month)?

19 Why was the capture of Tobruk important to the German Army?

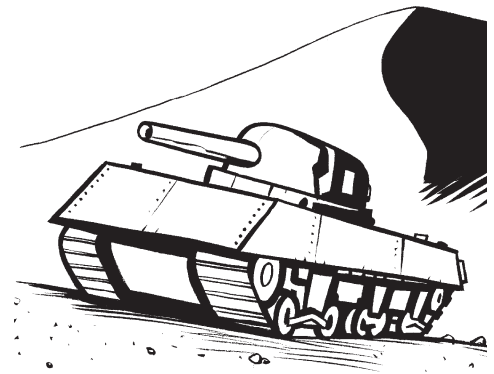
During their recreation time, the Australians liked to play cricket against the British. One special rule read 'Play to be continuous until 1800 hours, except by interference by air raids. Play will NOT, rpt NOT cease during shell fire.'

20 Change these times to 24-hour (or army) time.

- a** 12.30 pm **b** 2.30 am **c** 5.15 pm **d** 1.30 am

21 Change these army times to 12-hour time.

- a** 1800 **b** 0230 **c** 2300 **d** 1230

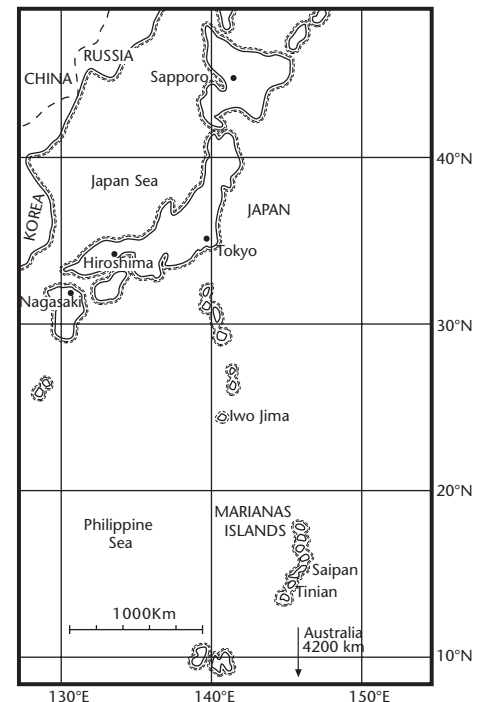




During the final stages of World War 2, the United States dropped two atomic bombs over the Japanese cities of Hiroshima and Nagasaki in a successful attempt to hasten the end of the war. The nuclear weapon known as 'Little Boy' was dropped over Hiroshima on August 6 1945. 'Fat Man' was released over Nagasaki on August 9 of the same year.

The capture of Tinian, one of the Marianas Islands, provided American forces with a valuable airstrip from which to mount raids on Japan. The islands served as the base from which the atomic bomb attacks on Hiroshima and Nagasaki were staged.

- 1 Colour the seas, nations and islands different colours.
- 2 Provide an approximate latitude and longitude for:
a Tinian **b** Nagasaki **c** Sapporo.
- 3 Name the features at:
a 26°N, 141°E **b** 16°N, 146°E **c** 45°N, 129°E



Hiroshima was the primary target of the first nuclear bombing mission on August 6. A B-29 bomber was launched from Tinian in the west Pacific, about six hours flight time from Japan. Two other B-29s were also launched. After leaving Tinian the aircraft made their way separately to Iwo Jima where they met at an altitude of 2440 metres and set course for Japan. The aircraft arrived over the target in clear visibility at 9855 metres.

- 4 On the map, draw the course the B-29s took to fly to Japan.
- 5 Approximately how far did the B-29s travel in kilometres? Don't forget to include the return trip.
- 6 Calculate the average speed of the B-29s for their flight to Hiroshima.
- 7 In kilometres, what was the altitude of the bombers when they were at:
a Iwo Jima? **b** Hiroshima?
- 8 How much higher were the bombers at Hiroshima than Iwo Jima?
- 9 Why do think the bombers increased their altitude at Hiroshima?

The gravity bomb known as 'Little Boy' was released at 8.15 am from a height of 9855 metres. 'Little Boy' contained 60 kilograms of uranium-235 and took 57 seconds to fall from the aircraft to the predetermined detonation height about 600 m above the city. It created a blast equivalent to about 13 kilotons of TNT. The uranium-235 weapon was considered very inefficient, with only 1.38% of its material exploding.

- 10 Using the 24-hour clock, what time was the bomb dropped?
- 11 How many tonnes of uranium-235 were contained in the bomb?



12 How many minutes did it take for the bomb to detonate once it was released?

13 Calculate the average speed of descent of the bomb.

14 How many grams of uranium-235:

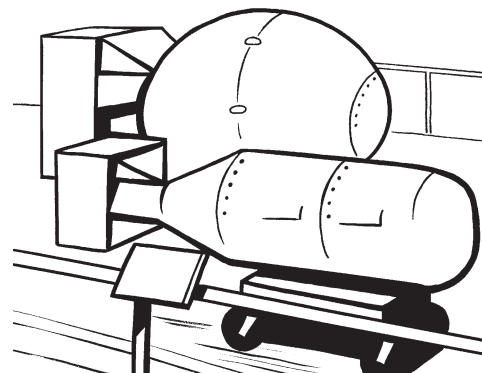
- a** fissioned (exploded)? **b** did not explode?

The radius of total destruction was about 1.6 km, with resulting fires across 11 km².

15 What was the total area of destruction?

16 How many square metres is this?

17 Calculate the diameter, in metres, of the space that contained the fires (assume a circular shape).



At the time of the attack the population of Hiroshima was approximately 345 000.

Approximately 80 000 people were killed immediately and another 70 000 were injured. It has been estimated that, of the people who died on the day of the explosion, 60% died from burns, 30% from falling bricks, timber and other materials and 10% from other causes.

18 Find the total number of people who were killed or injured by the blast.

19 Of the number of people who were killed or injured, what percentage was killed immediately by the blast?

20 Of the people who were killed in the blast, how many died from:

- a** flash or flame burns? **b** falling debris? **c** other causes?

On the morning of August 9 1945, the nuclear bomb code-named Fat Man was detonated. The bomb contained 6.4 kg of plutonium-239, and was dropped on Nagasaki's industrial valley at 11.01 am. The estimated population of Nagasaki at the time of the bombing was 270 000. Fat Man exploded 43 seconds later at 469 m above the ground.

21 What was the time when Fat Man exploded?

22 Using the 24-hour clock, what time was Fat Man dropped?

23 Of the total population of Nagasaki, calculate the percentage of immediate deaths if 60 000 people were killed in the blast.

24 How many grams of plutonium-239 were contained in 'Fat Man'?

On August 15, six days after the detonation over Nagasaki, Japan announced its surrender to the Allied Powers. They signed the Instrument of Surrender on September 2, officially ending the Pacific War and therefore World War 2.

25 After the bombing of Nagasaki, how many hours did it take for the Japanese to surrender?



In World War 2, Australian women fulfilled similar roles to those of women in World War 1. For most of this war, except for the Australian Women's Army Service (AWAS), nurses were the only females to serve outside of Australia in any capacity.

During World War 2 approximately 3500 nurses served in the Australian forces and 70 never returned, losing their lives during active service overseas.

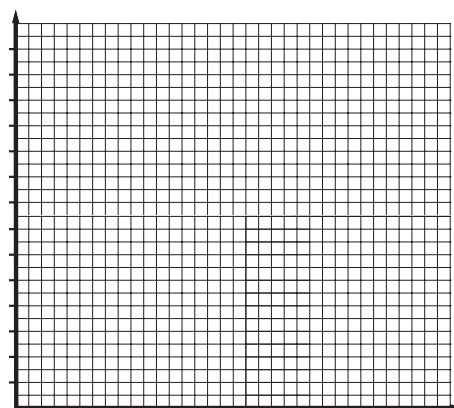
- 1 What do you think were the major causes of the nurses dying while on overseas duty?
- 2 What percentage of the nurses died on duty while overseas?

In World War 2, 691 000 men served in the Army, 45 800 men served in the Royal Australian Navy and 189 700 men served in the Royal Australian Air Force.

- 3 Find the total number of men serving in the Australian defence forces.

- 4 Complete the table below which shows the numbers of women participating in various roles in the Australian defence forces during World War 2.

	Service	Medical/nurses	Auxiliary	Total
a	Army	11 500	24 000	
b	Navy	60	3 000	
c	Air Force	600	27 000	
d	Total			



- 5 Draw a column graph to compare the participation rates in each service.
- 6 What was the total number of women participating?
- 7 Calculate the percentage of women making up the:
 - a Army
 - b Navy
 - c Air Force.
- 8 Of the women in each service, what is the fraction, in simplest form, of medical/nurses in the:
 - a Army?
 - b Navy?
 - c Air Force?
- 9 Of the women in each service, what is the percentage of auxiliary women in the:
 - a Army?
 - b Navy?
 - c Air Force?

The number of women in the Australian workforce for a range of occupations is shown in the graph over the page.

- 10 List the five main areas of employment for women in 1910.

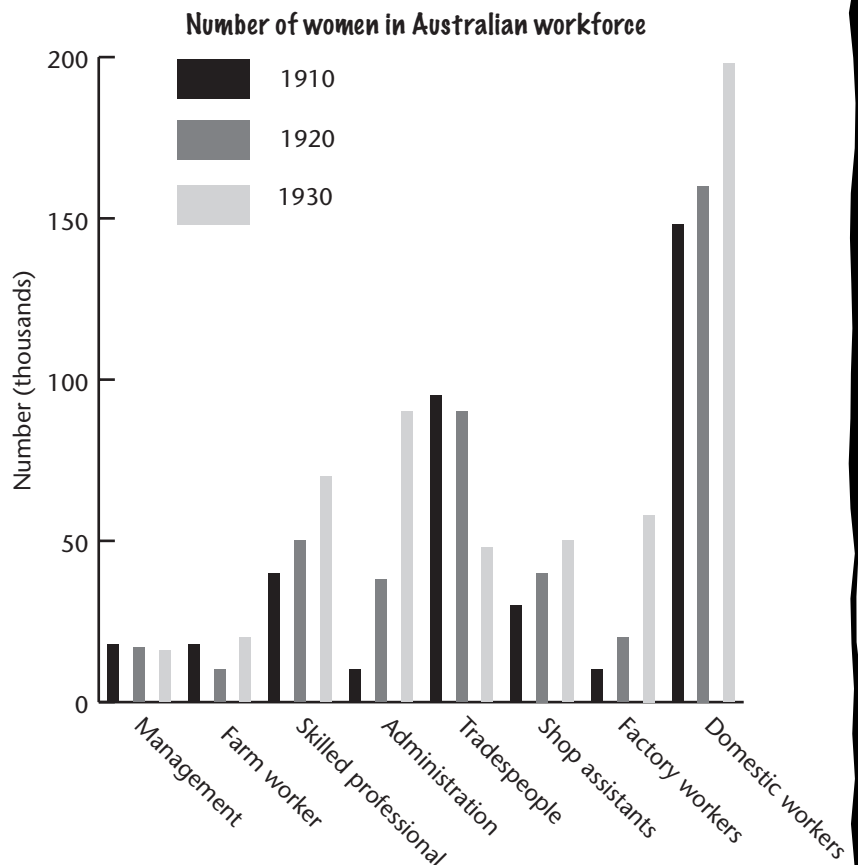


The changing roles of women in war (cont'd)



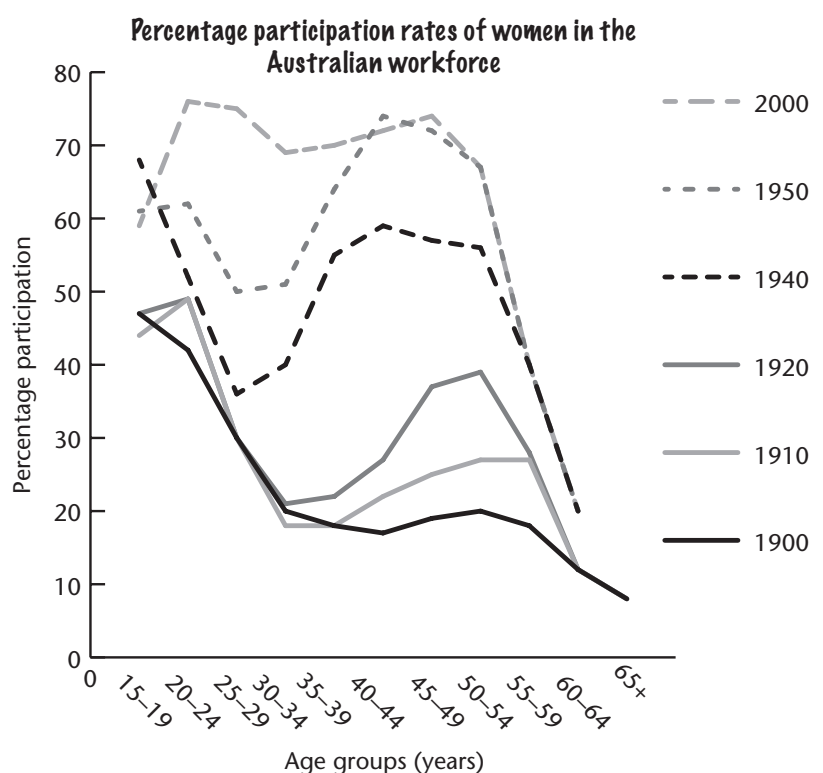
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- 11 The recorded female population of Australia in 1910 was 2 155 770. What percentage of these women were employed as domestic workers?
- 12 List the five main areas of employment for women in 1920.
- 13 The recorded female population of Australia in 1920 was 2 683 229. What percentage of these women were employed as domestic workers?
- 14 The recorded female population of Australia in 1930 was 3 262 728. What percentage of these women were employed as domestic workers?
- 15 Do you think World War 1 changed the role of women in Australia? Explain.
- 16 Why do you think there was a large decrease in the Tradespeople group?



During the war there was a shortage of men to work in industry; for example to drive buses and to operate machinery. Women took on many of these roles.

- 17 The graph shows the participation rates of women in the Australian workforce from 1900 to 2000.
 - a What was the 40–44 year old participation rate in 1950?
 - b What was the 50–54 year old participation rate in 1920?
 - c What happened to the female participation rates following the end of the wars in 1918 and 1945?
 - d What happened to the female participation rates between 1950 and 2000?



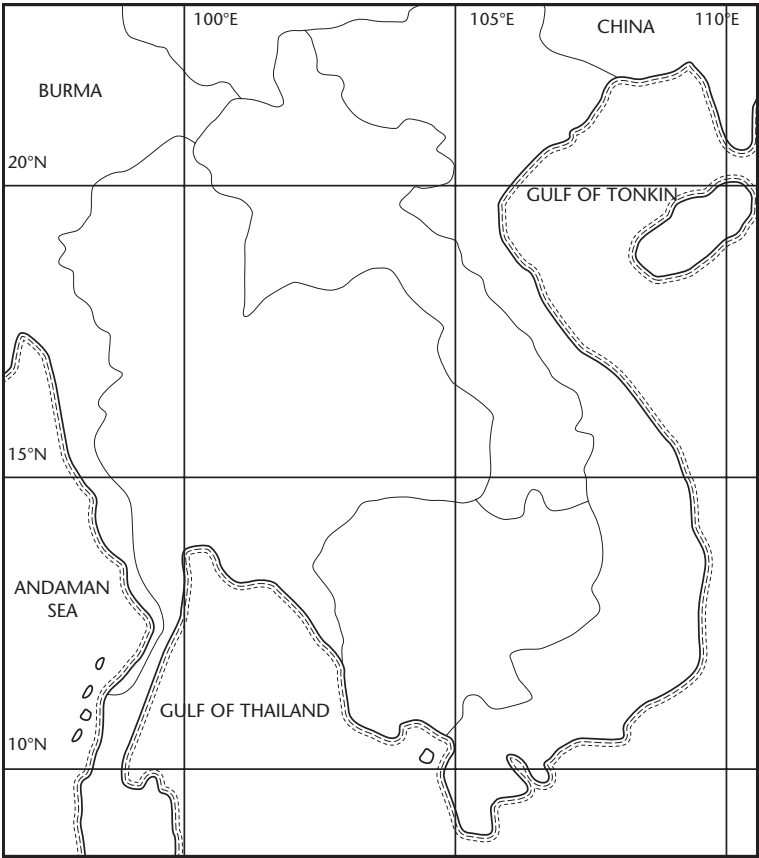
The Vietnam War



11

The Vietnam War was the longest major conflict Australian troops have been involved in. They entered the war in 1965.

- 1 Colour the seas on the map.
- 2 Label each of these nations and colour them.
 - a North Vietnam 20°N, 105°E
 - b South Vietnam 12°N, 108°E
 - c Cambodia 12°N, 105°E
 - d Thailand 15°N, 100°E
 - e Laos 22°N, 102°E
- 3 Label these locations on the map.
 - a Long Tan 10°33'N, 107°15'E
 - b Hanoi 21°2'N, 105°51'E
 - c Ho Chi Min City 10°46'N, 106°40'E
 - d Nui Dat 10°40'N, 107°40'E



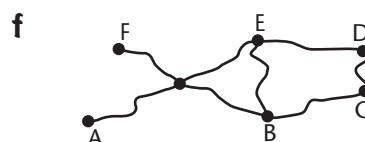
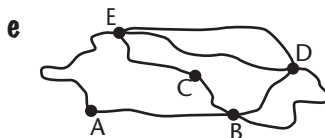
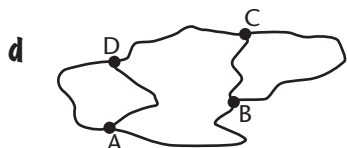
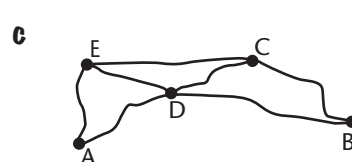
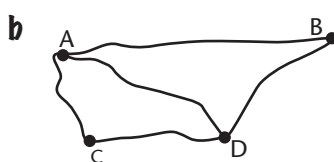
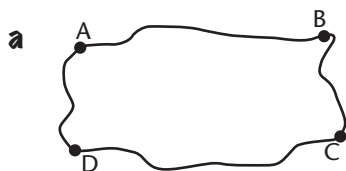
- 4 Complete the table below to find the total numbers of participants and casualties in the Vietnam War.

Nation	Size of force	Casualties	Nation	Size of force	Casualties
South Vietnam	650 000	220 400	North Vietnam	340 000	1 176 000
United States	536 000	58 160	China	170 000	1 450
South Korea	313 000	4 960	Soviet Union	3 000	16
Australia	50 000	520	North Korea	300	0
Thailand/ Philippines	10 500	1 350	Viet Cong	?	250 000
New Zealand	3 900	37			
Total	a	b	Total	c	d
Civilians, approximately 4 000 000					



In December 1965, the Viet Cong adopted guerrilla warfare tactics. They made hit and run attacks and ambushes. Complex systems of tunnels and caves were built in many areas which formed fighting bases and provided support for the troops. These tunnel complexes, often several stories deep, housed up to 10 000 people who lived underground for many years.

- 5** The tunnel networks below consist of large caves for living, storing food and ammunition. The caves are connected by various tunnels. Find out, for which tunnel networks it is possible to crawl through all the tunnels without going through any tunnel section more than once. Which networks can you start and finish at the same cave? Complete the table

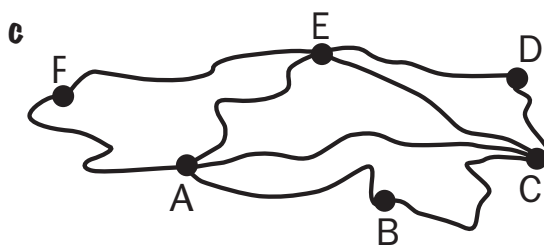
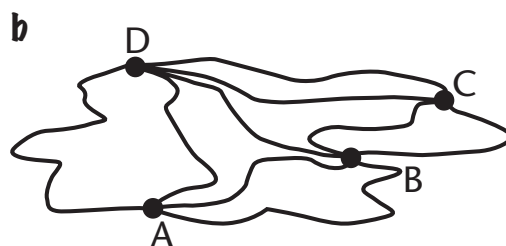
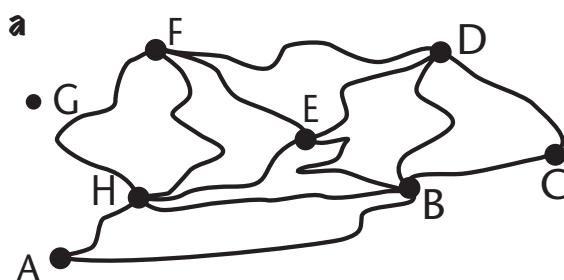


	No. of tunnels connecting each cave	Once only through each tunnel	Back to start cave
a	A=2, B=2.....	Yes	Yes
b			
c			
d	A=3,		
e			
f			

- 6** Describe any rules you notice to help decide if you can travel through all the tunnels once and return to:

- a** the cave you started from **b** another cave.

- 7** Test your rules on these cave networks.





The Australian government supported the invasion of Iraq by United States forces to search for nuclear, biological or chemical weapons. The Australian government supported the view that ousting Saddam Hussein from power was the right thing to do.



1 Complete the table showing the estimated size of force from participating nations.

	Nation	Population of nation	Size of force	Force as a % of population
a	United States of America	280 million	214 000	
b	United Kingdom	60 million	45 000	
c	Australia	20 million	2 000	
d	Iraq	23 million	400 000	

2 Which nations contributed most and least, in terms of population?

3 The table below shows an approximate breakdown of the amount of money Australia has spent on the Iraq war. On the axes provided, draw a column graph of the data.

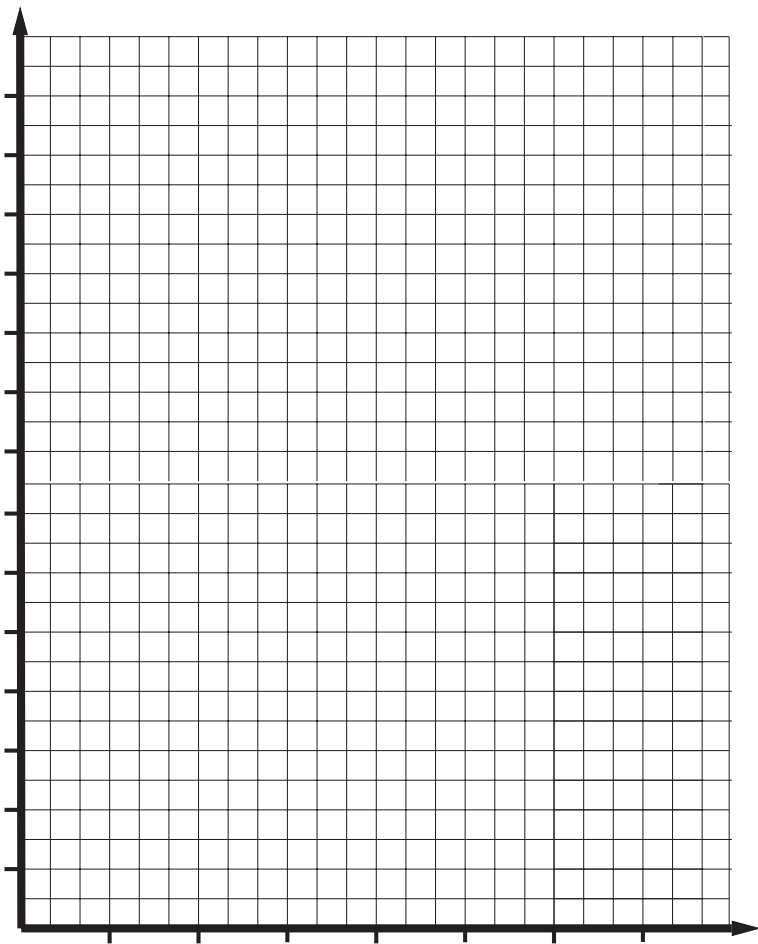
4 Explain the shape of the graph.

5 According to these figures, calculate the total amount Australia has spent on the war so far.

6 Write this amount using a numeral.

7 If the population of Australia is 21.9 million people, how much has been spent per person?

8 How much is this per person per year?



Year	2003	2004	2005	2006	2007	2008	2009	2010
Amount (£ million)	112	158	178	210	272	293	198	135



- 9 Complete the table showing the breakdown of 4773 troop casualties in the Iraq war.

	Country	Number of casualties	Proportion %		Country	Number of casualties	Proportion %
a	Australia	21		i	Latvia	3	
b	Bulgaria	13		j	Romania		0.06
c	Denmark		0.15	k	Slovakia		0.08
d	El Salvador		0.11	l	Ukraine	18	
e	Georgia	5		m	Thailand		0.04
f	Italy	33		n	USA	4404	
g	Poland		0.63	o	Spain	11	
h	United Kingdom	179		p	Others	9	

- 10 Locate each country on the map and colour it.

- a Saudi Arabia 30°N40°E b Syria 35°N40°E
c Turkey 38°N40°E d Kuwait 28°N47°E

- 11 Locate these cities and label them.

- a Baghdad 33°N44°E
b Kuwait City 29°N48°E
c Basra 31°N48°E
d Mosul 36°N43°E

- 12 Calculate the distance from:

- a Basra to Baghdad
b Kuwait City to Baghdad
c Kuwait City to Basra
d Baghdad to Mosul.

- 13 Find the true bearing from:

- a Baghdad to Kuwait City
b Kuwait City to Basra
c Kuwait City to Baghdad
d Mosul to Baghdad.

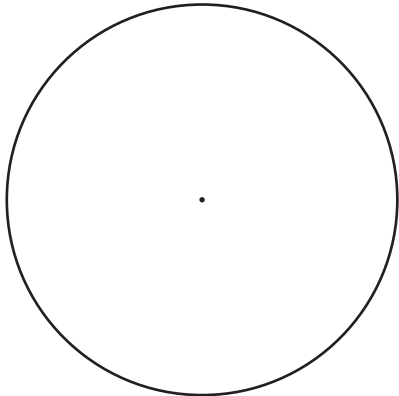




The damage to the infrastructure of a country is one of the least publicised casualties of war. Billions of dollars in grants have been made to rebuild schools, water supplies, sewer systems, banks and hospitals in Iraq.

1 Complete the table showing the estimated costs to reconstruct Iraq. Draw a sector graph.

	Item	Amount (£millions)	Sector angle
a	Government operations	781	
b	Essential services	22 310	
c	Humanitarian services	3 700	
d	Health and education	6 410	
e	Regional programs	3 030	
f	Governance	390	
g	Economy	420	
h	Buildings reconstruction	1 830	

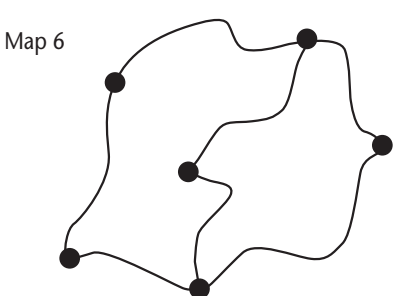
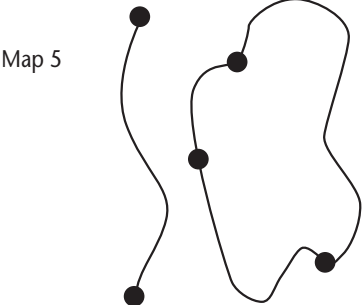
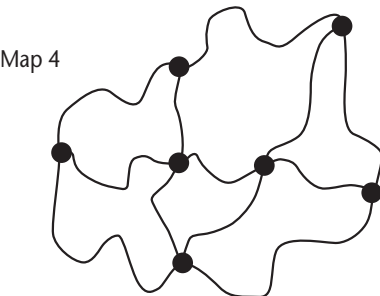
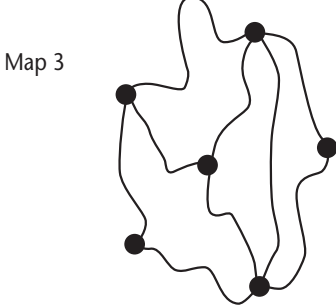
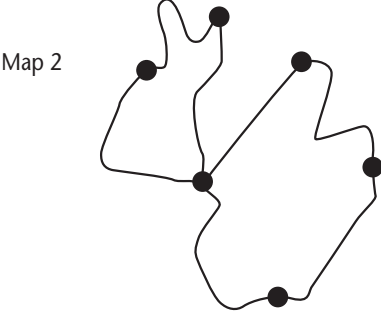
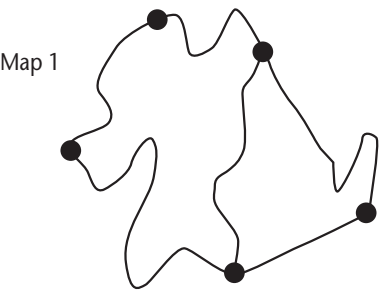


2 Calculate the total amount, in billions of dollars, needed in the estimates to reconstruct Iraq.

3 On the road maps below the dots represent towns and cities. Engineers need to visit each town on the map to reconstruct buildings and services in each town. On which maps can you visit each town once only and:

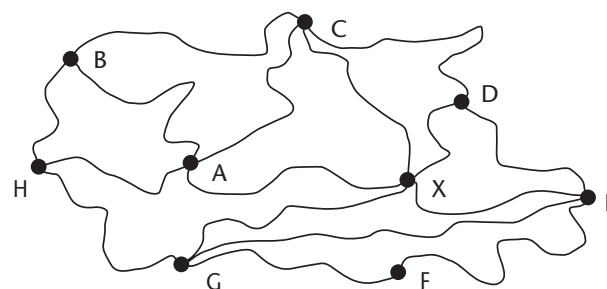
a return to the starting point?

b finish at another point?





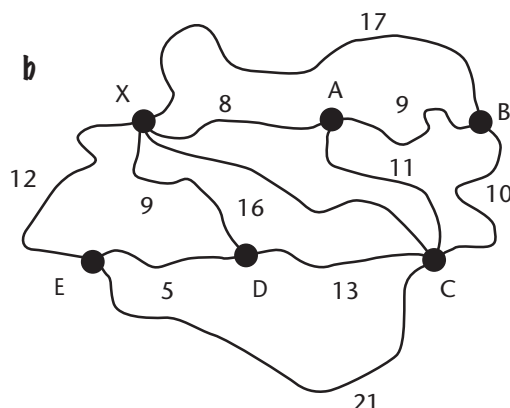
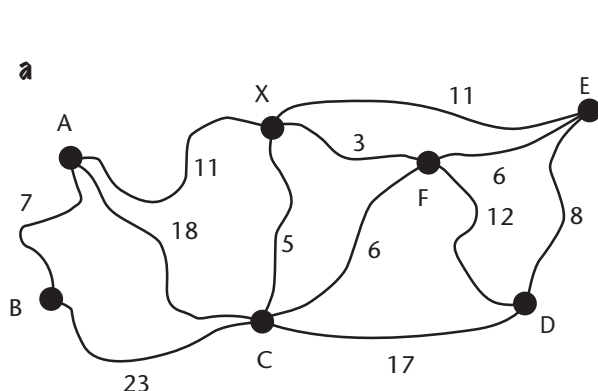
- 4** A food delivery van leaves the depot at town X on the map. The vehicle must deliver food to each town and return to the depot. Plan a delivery route for the driver so that he visits each town only once.



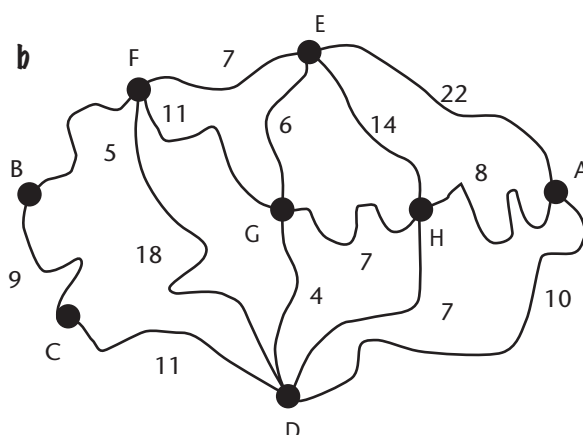
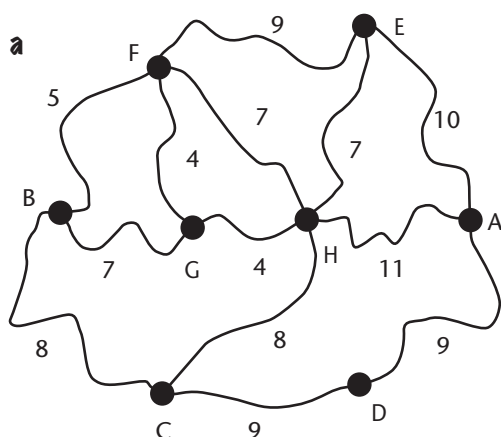
- 5** Plan a delivery route leaving from E and returning to E.

- 6** Can this be done from all towns?

- 7** A large water storage is located at X. Find the shortest length of piping needed to provide each town with water in each of the network maps below.



- 8** A water tanker has to deliver water to the towns on each map. The vehicle leaves the depot at town A. Find the shortest route the vehicle can take to visit each town once only.



- 9** A power station is built at each town A in Q7. Find the shortest length of power line to connect power to the towns without forming a loop.

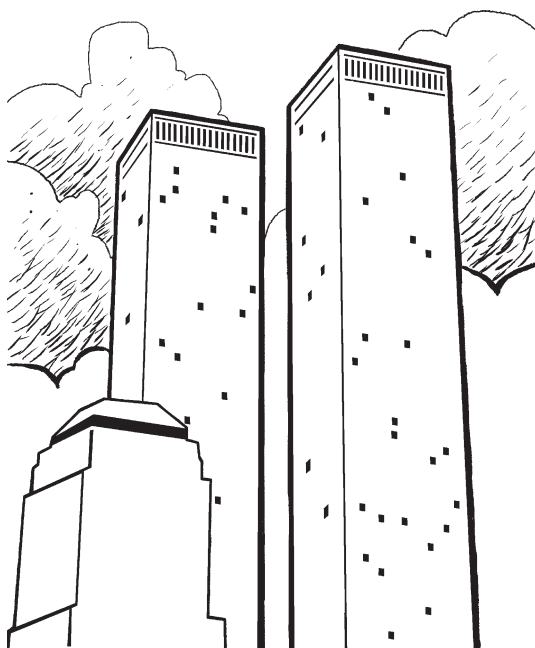
- 10** If it costs \$56 000 per km for power lines, find the cost of construction of both the power lines in Q7.



The War on Terror was launched following the terrorist attacks on the United States on September 11 2001. It is a war against any terrorist activities or organisations and is supported by the United States, United Kingdom and many other countries.

The World Trade Center was constructed in the 1960s. The two main towers had 110 stories and were 410 m high, with 920 m² of office space. On completion in 1972 the North Tower became the tallest building in the world, a record it held for two years. It stood 417 m tall and featured a 110 m telecommunications antenna at the top, making the tower 527 m tall.

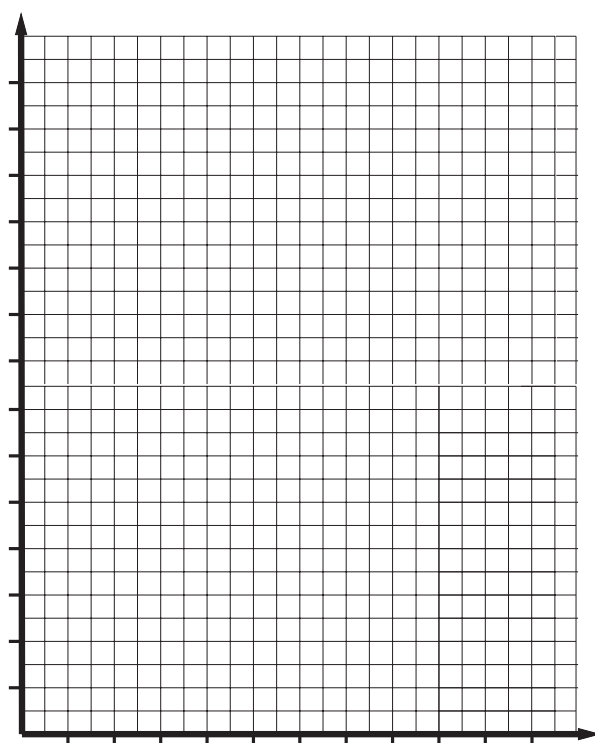
On completion in 1973 the South Tower became the second tallest building in the world. Its rooftop was 420 m high. Each tower was 63.4 m² at the base. The Leaning Tower of Pisa stands at 55.86 m and has a diameter of 15.5 m.



- 1 Draw a column graph on the grid below to compare the overall heights and widths of the towers with the Leaning Tower of Pisa.
- 2 Both the North and South Towers are 110 stories high. The first aeroplane hit the North Tower between the 93rd and 99th stories. The second aeroplane hit the South Tower between the 77th and the 85th stories.

Mark these positions on your column graph.

- 3 Each tower contained 196 000 tonnes of steel. How many kilograms is this?
- 4 Each tower weighed 510 000 tonnes. How many kilograms is this?
- 5 Each tower contained 324 000 cubic metres of concrete. If one cubic metre weighs 2.4 tonnes, how many tonnes of concrete were used in each tower?
- 6 Each tower had 43 600 windows. How many windows in total were in both towers?
- 7 Each tower used 19 300 km of electric cabling.
 - a How many metres is this?
 - b It is approximately 400 m around an athletics track. How many times would the cable used in the towers go around the track?

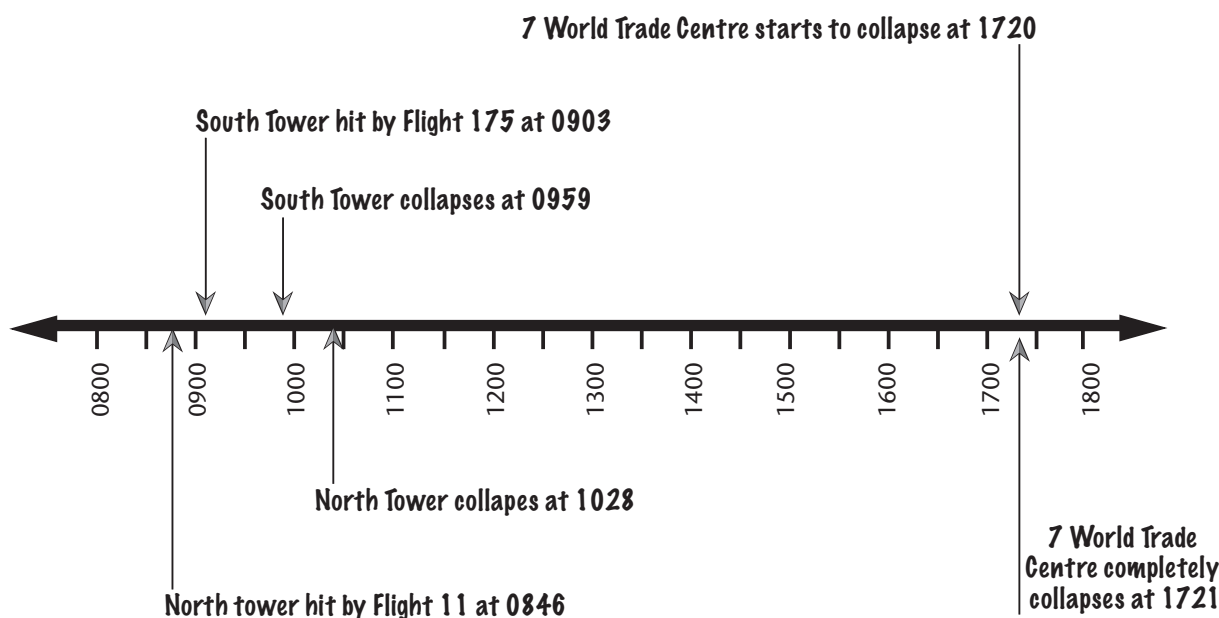


The war on terror begins (cont'd)



14

The North and South Tower collapsed as a result of the damage from the attacks. The debris from the North Tower ignited fires in the building across the road. The building, known as 7 World Trade Center, eventually collapsed. The following timeline shows this sequence of events.



- 8 Convert these times using the 12-hour clock.

a 0846	b 0903	c 0959
d 1028	e 1720	f 1721
 - 9 How long after being struck did the North Tower collapse? Give your answer in:

a minutes	b seconds	c hours.
-----------	-----------	----------
 - 10 How long after being struck did the South Tower collapse? Give your answer in:

a minutes	b seconds	c hours.
-----------	-----------	----------
 - 11 How long did it take after the initial impact on the North Tower for all the buildings in and around the World Trade Center to collapse? Give your answer in minutes.
- The terrorist attacks, and the subsequent four-day grounding of aircraft immediately following, had many consequences for the aviation industry around the world. Airlines that were already in financial trouble experienced further economic problems as a result.*

Blue Skies Air is an Australian airline with both domestic and international routes.

- 12 On July 11 2001 the airline reported a share price of \$48. Leading up to the terrorist attacks the share price fell 5% per month. What was the share price at the time of the attacks?
- 13 Immediately after the attacks the share price fell by a further 18% percent. What was the new share price?



Military statistics, especially those related to death and injury, are particularly difficult to calculate. There are also problems ascertaining the accuracy of any data that is gathered in areas where infrastructure has been damaged in conflict areas.

1 Complete the tables below.

	War	Years	Enlisted	Deaths	Wounded	Captured
a	Sudan	1860–1861	770	9	0	0
b	South Africa	1899–1902	16 000	518	538	100
c	Boxer Rebellion	1900–1901	560	6	0	0
d	World War 1	1914–1918	417 000	60 000		4 044
e	World War 2	1939–1945	1 018 800	35 000	66 553	30 700
f	Korean War	1950–1953	15 164		1 216	29
g	Malayan Emergency	1950–1960	7 000	36	20	0
h	Indonesian Confrontation	1963–1966	3 500	15	0	0
i	Vietnam War	1962–1972	50 001	496		0

	War	Total casualties and captured	Percentage of enlisted who were casualties or captured
a	Sudan	$9 + 0 + 0 = 9$	$\frac{9}{770} \times 100 = 1.17\%$
b	South Africa		
c	Boxer Rebellion		
d	World War 1	219 044	
e	World War 2		
f	Korean War	1 584	
g	Malayan Emergency		
h	Indonesian Confrontation		
i	Vietnam War	2 894	

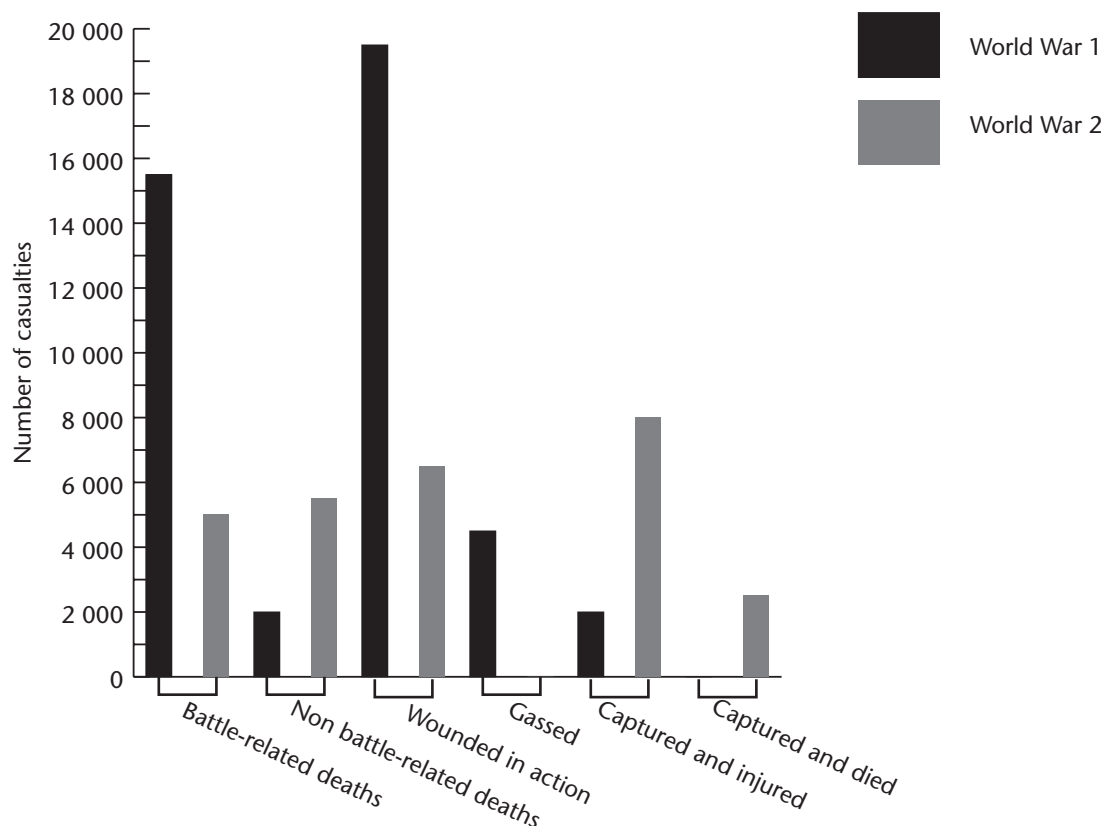
- Which war do you think was worst for the Australian defence forces? Why?
- Draw a timeline to show Australia's participation in various wars.
- The population of Australia in 1914 (World War 1) was approximately 4 million people and in 1939 (World War 2) it was about 6.9 million. Calculate the number of deaths, captures and injuries for each world war as a percentage of the population at the time.





A local historian wanted to gather data on the casualties of World War 1 and 2 in the state where he lives. He concluded that although the data he gathered was only an estimation, it would still be useful for comparing the location and type of injury sustained by defence forces who lived in this state. Below is the column graph he used to display his findings. State all answers correct to the nearest unit.

Estimate of casualties sustained in World War 1 and 2 by occupants of one Australian state



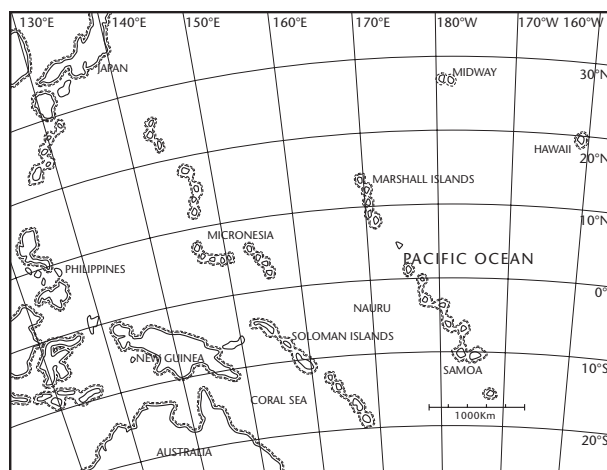
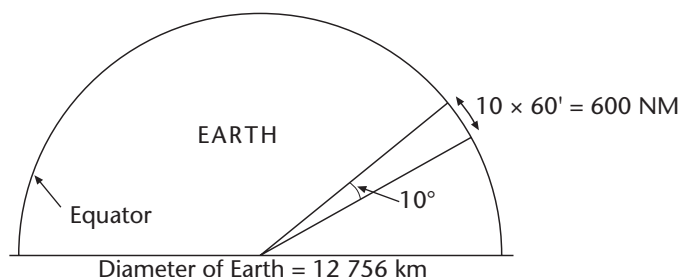
- 5 What is the total number of casualties reported in this state:
 - a in World War 1?
 - b in World War 2?
 - c for both wars?
- 6 What percentage of the total casualties from World War 2:
 - a were wounded in action?
 - b were captured and injured?
 - c died in battle?
 - d died in non-battle related incidents?
- 7 Calculate the ratio of casualties:
 - a wounded in action in World War 1 to World War 2
 - b gassed in World War 1 to those who were captured and injured in the same war.
- 8 From the total number of casualties reported for both wars, what percentage were wounded in action in World War 2?
- 9 Compare the data in the historian's graph to the data for World War 1 and 2 in the table on the previous page. Do you think the death and injuries sustained by the defence forces who lived in this state are representative of the whole of Australia? List some reasons for any variation in the data.



World War 1 was largely fought and won on land but this could not have taken place without the movement of ships. Ships were used to transport supplies and manpower needed on the battlefields from overseas countries. Command of the seas and supply lines was essential to the armed forces on land.

The nautical mile is based on the circumference of the planet Earth. If you divide the circle made by the equator into 360 degrees and then divide a degree into 60 minutes, a single minute of arc on the planet Earth is 1 nautical mile. The nautical mile NM is a non-SI unit used mainly by navigators in the shipping and aviation industries.

- 1 How many degrees is a full rotation around the equator of the Earth?
- 2 How many minutes is this?
- 3 How many nautical miles is this?
- 4 If the diameter of the Earth is 12 576 km at the equator, calculate the circumference of the Earth at the equator using the rule $C = \pi \times D$. (use $\pi = 3.14$)
- 5 Find in kilometres the length of the nautical mile.
- 6 Use the map to find the distance in nautical miles a ship would have to travel: (Hint: Find the degrees travelled and convert to NM)
 - a due west from Hawaii to the Micronesia Islands
 - b due south from Japan to New Guinea
 - c due south from Midway then due west to Nauru
 - d due south from Hawaii then due west to the Coral Sea
 - e due north from Samoa to Midway.



The device that sailors used to make their measurement of speed was a rope with knots tied at equal distances along it, tied to a chip of wood. The sailor would throw the chip of wood over the side and when the chip was in the water they would start to count the number of knots that passed over the back of the ship in thirty seconds. This was the speed the ship was travelling. The knot is defined as one nautical mile per hour.

- 7 How fast is a ship travelling if:
 - a 10 knots pass in 30 seconds?
 - b 15 knots pass in one minute?
- 8 A warship can cruise at 35 knots. How long will it take to travel:
 - a 35 NM?
 - b 1000 NM?
 - c 1200 NM?

-

- | Position | Distance to travel (NM) | Time to cruise to the scene |
|-----------------------|-------------------------|-----------------------------|
| Port Moresby 9°S147°E | | |
| Nauru Island 2°S166°E | | |
| Townsville 19°S146°E | | |
| Micronesia 10°N150°E | | |
| Samoa 10°S175°W | | |

- | Date | Position | Distance travelled from last position | Time taken | Places of interest |
|-----------|--------------------------|---------------------------------------|------------|--------------------|
| 1 January | Off Hawaii 21°N158°W | 0 | | |
| January | 10°S160°W | | | |
| January | 10°S175°W | | | |
| | 10°S166°E then 1°S166°E | | | |
| | 1°S150°E then 10°N150°E | | | |
| | 10°N178°W then 28°N178°W | | | |

- 35



The tank is an armoured fighting vehicle designed for frontline combat over any terrain. The name 'tank' came as a result of trying to hide its purpose during construction, where it was labelled as a water carrier or water tank. The first tank to engage in battle was during the Battle of Flers-Courcellette in September 1916.

- 1 Locate the position of (5, 2) on the grid as your starting point. Connect the points to form a picture.

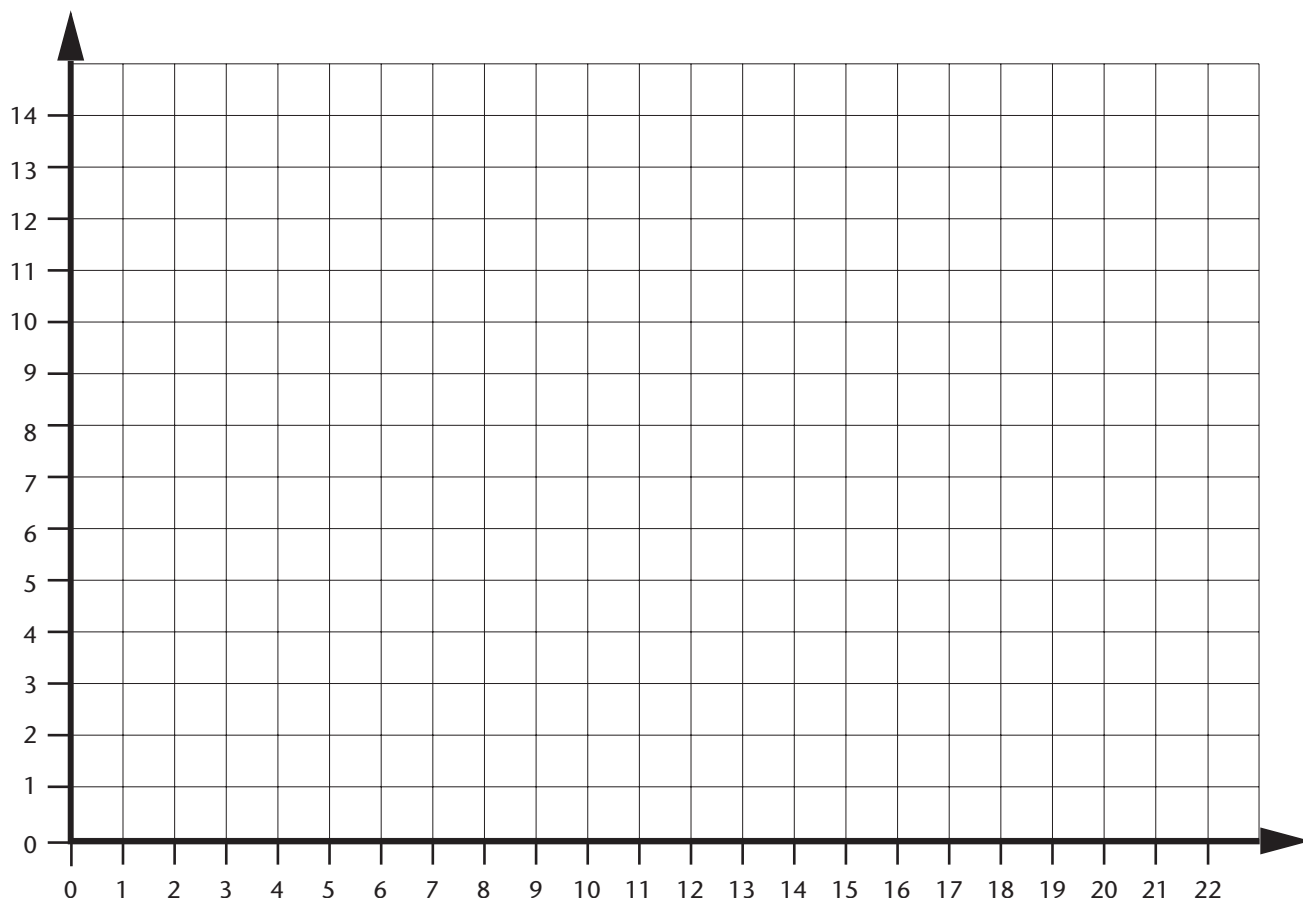
Start: (5, 2) (8, 2) (11, 2) (15, 2) (18, 3) (19, 4) (19, 5) (19.5, 6) (19, 7) (16, 7.5) (14, 8) (13, 11) (10, 11) (10, 8) (6, 8) (5.5, 7) (2, 7) (1.5, 6) (1.5, 5.5). Start: (10, 11) (8, 10.5) (8, 9) (7.5, 8) (7, 8.5) (6, 8.5) (6, 8.75) (1, 8.75) (1, 9.25) (6, 9.25) (6, 8.75). Start: (6, 9.25) (6, 9.5) (7, 9.5) (8, 10.5) (10, 11). Start: (8.5, 9) (9.5, 9) (9.5, 10) (8.5, 10) (8.5, 9). Start: (10.5, 8.5) (12.5, 8.5) (12.5, 10.5) (10.5, 10.5) (10.5, 8.5). Start: (1.5, 5.5) (1.75, 6) (2, 6.75) (5.5, 6.75) (10, 6.75) (15, 6.75) (19, 6.5) (19.25, 6) (19, 5). Start: (5, 2) (2, 3) (1, 4) (1.25, 5) (1.5, 5.5) (2, 6) (6, 6.25) (10.5, 6.5) (14, 6.25) (17.5, 6.25) (18.5, 5.5) (19, 5). Start: (5.75, 7.75) (3.5, 7.75) (3.5, 7.25) (5.5, 7.5). Start: (6.5, 7.75) (6.5, 7.25) (7.5, 7.25) (7.5, 7.75) (6.5, 7.75). Start: (7, 8.5) (7, 9.5). Start: (5.5, 7) (12, 7) (19, 7).

Draw circles of radius of 1 unit at these positions: (5, 3) (7, 3) (9, 3) (11, 3) (13, 3) and (15, 3)

Draw circles of radius of 1.5 units at these positions: (2.5, 4.5) and (17.5, 4.5)

Draw circles of radius of 0.5 units at these positions: (6, 5.75) (10.5, 6) and (14, 5.75)

Complete the picture by adding soldiers and gunners: Colour the picture.





The main cannon in a tank is normally contained in a rotating turret with secondary machine guns. The main battle tank is highly mobile and able to travel over most types of terrain due to its continuous tracks. A tank can travel at approximately 40 km/h across rough terrain and up to 70 km/h on roads.

- 2 Calculate the time taken, in hours and minutes for a tank to travel these distances over rough terrain.

a 80 km	b 120 km	c 200 km	d 350 km
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- 3 If the speed of a tank is 35 km/h how far can it travel in:

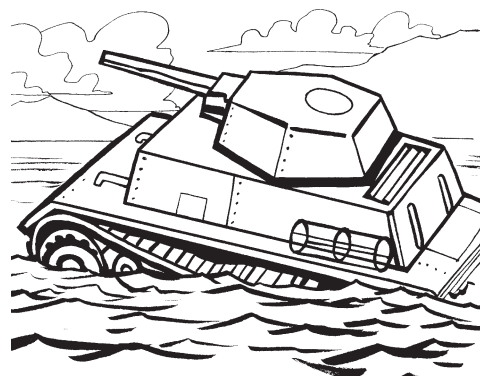
a 3 h?	b 5 h?	c 2.6 h?	d 2 h 30 min?
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In the absence of any bridges, most tanks must cross a river. The typical safe depth for tanks is approximately 1 m, being limited by the height of the engines air intake and driver's position. Some modern tanks can cross rivers up to a depth of 3.5 m when properly prepared and equipped with a snorkel to supply air for the engine and crew.

- 4 Convert these river depths to centimetres.

a 3 m	b 2 m
c 2 m 45 cm	d 3.54 m
- 5 Can a modern tank cross a river that has a depth of:

a 80 cm?	b 520 cm?
c 256 cm?	d 350 m?

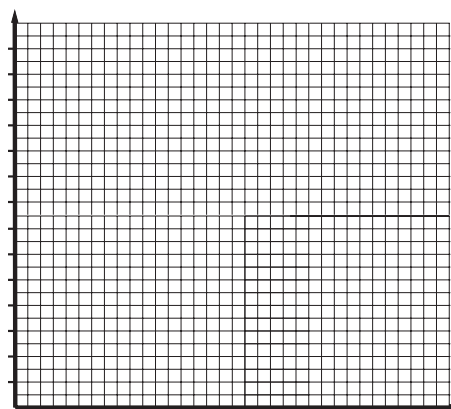


- 6 Why do you think tank crews may feel strange crossing deep rivers?
- 7 What do you think are some advantages of tanks being able to cross deeper rivers?
- 8 The range of a particular tank is 450 km by road and 250 km cross-country. Calculate the fuel consumption of a tank in L/100 km if the fuel capacity of the tank is 700 L when the tank is travelling:

a on the road	b cross-country.
---------------	------------------

c The fuel consumption of a car is approximately 10 L/100 km. Compare this to the tank.
- 9 The table compares the power output of a tank engine with that of a car and a train. On the axes below draw a column graph of the data in the table.

Vehicle	Engine power (hp)	Power/weight (hp/t)
Toyota Camry	160	110
Lamborghini sports car	630	380
Formula one racing car	950	2 100
Battle Tank	1 500	25
Locomotive	2 500	12





A guided missile is a self-propelled rocket used as a weapon. Missiles are propelled by a jet engine and generally have one or more explosive warheads. In military terms, the word missile describes a powered, guided munition, whilst the word rocket describes a powered, unguided munition.

A cruise missile is basically a small, pilotless airplane. Cruise missiles have a 2.6-metre wingspan, are powered by turbofan engines and can fly 800 to 1600 kilometres, depending on their design.

- 1 How many centimetres in:

a 2.61 m?	b 2.34 m?	c 2.5 m?	d 5.03 m?
-----------	-----------	----------	-----------
- 2 How many millimetres in:

a 2.612 m?	b 2.34 m?	c 2.3 m?	d 2.13 m?
------------	-----------	----------	-----------
- 3 How many metres in:

a 800 km?	b 1200 km?	c 1500 km?	d 1340.6 km?
-----------	------------	------------	--------------

The job of a cruise missile is to deliver a 450 kilogram high-explosive bomb to a precise location – the target. The missile is destroyed when the bomb explodes. Since cruise missiles cost between \$500 000 and \$1 000 000 each, it's a fairly expensive way to deliver a 450 kilogram package.

- 4 Calculate the median cost of one missile.
- 5 Use this price to calculate the cost, in \$millions, of firing:

a 3 missiles	b 6 missiles
c 12 missiles	d 23 missiles.
- 6 Calculate the weight in tonnes of explosive for:

a 2 missiles	b 6 missiles	c 16 missiles	d 34 missiles.
--------------	--------------	---------------	----------------
- 7 If a missile costs \$650 000, how many missiles can be purchased for:

a \$1 300 000?	b \$2 600 000?	c \$5 200 000?	d \$7 million?
----------------	----------------	----------------	----------------
- 8 Calculate the number of missiles needed to deliver these quantities of explosives.

a 900 kg	b 450 kg	c 2700 kg	d 4500 kg
----------	----------	-----------	-----------

Cruise missiles are 6 m long and 500 cm in diameter. At launch, they include a 250 kg solid rocket booster and weigh approximately 1500 kg.

- 9 Select a classmate and measure their:

a height in metres	b weight in kilograms.
--------------------	------------------------
- 10 How many students are needed to model these features of a cruise missile?

a length	b weight
----------	----------
- 11 Draw a diagram which shows the number of students needed to produce a model of a missile.



The booster falls away once it has burned its fuel. The wings, tail fins and air inlet unfold, and the turbofan engine takes over. This engine weighs just 65 kg and produces 270 kg of thrust burning RJ4 fuel. The 600 L of fuel weighs approximately 450 kg at launch.

12 How many grams in:

- a** 23 kg? **b** 45 kg? **c** 64.6 kg? **d** 450 kg 100 g?

13 How many kilograms in:

- a** 43 000 g? **b** 12 000 g? **c** 650 100 g? **d** 400 600 g?

14 The petrol tank in some large four wheel drive vehicles can hold 65 litres of fuel. How many of these car tanks would be needed to launch the missile?

15 How many millilitres in:

- a** 40 L?
b 600 L?
c 500.6 L?
d 650.2 L?

16 How many litres in:

- a** 6 000 mL?
b 320 000 mL?
c 610 800 mL?
d 563 200 mL?

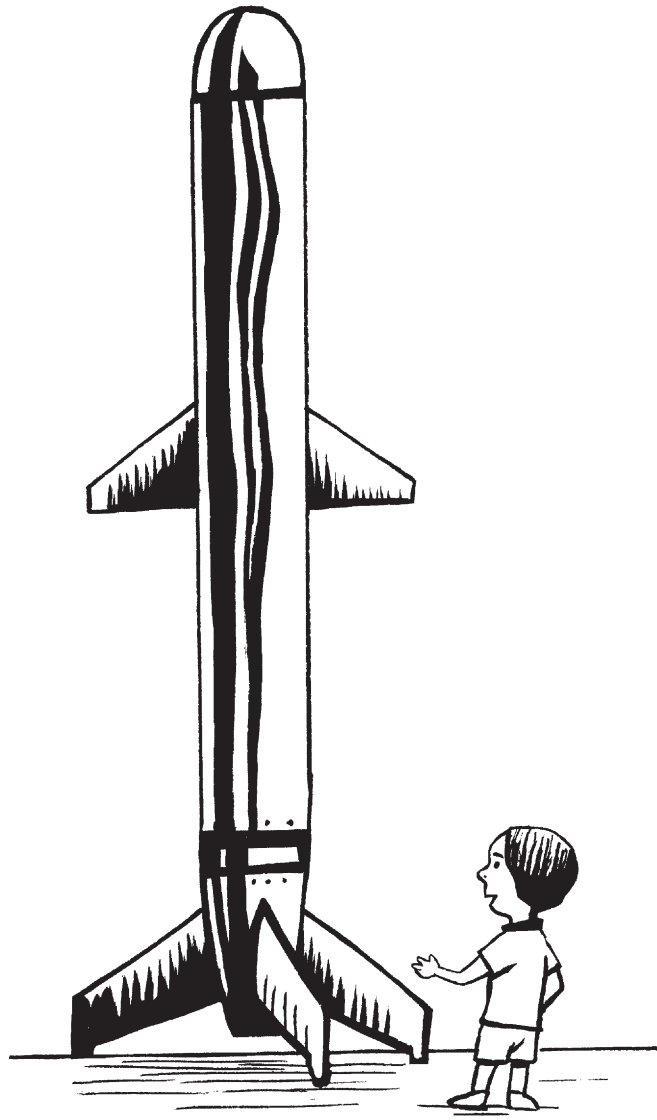
The missile has a cruising speed of about 880 km/h.

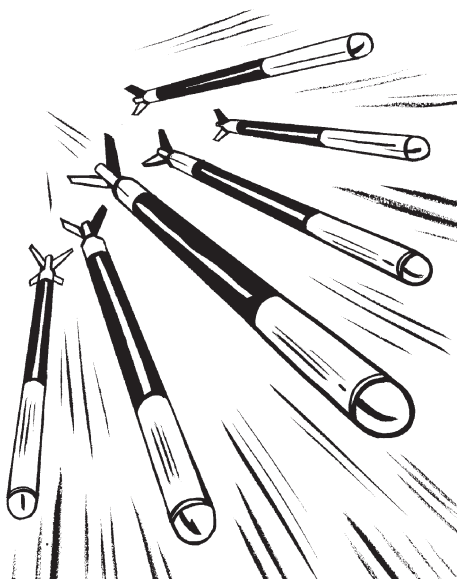
17 How far can the missile travel in:

- a** 2 h?
b 3 h?
c 3.5 h?
d 2h 30 m?

18 Calculate the flying speed of each of these missiles, if it travels:

- a** 1200 km in 2 hours
b 2600 km in 3 hours
c 3520 km in 4 hours
d 7500 km in 6 h 20 min.





A ballistic missile is a munition that follows a parabolic flight path; it is a guided missile. Stinger missiles can be launched by foot soldiers. To fire the weapon, the soldier aims the missile at the target and the seeker locks on. Upon firing, a small launch rocket shoots the missile out of the launch tube. The launch engine falls away and the main solid rocket engine lights. The missile then flies to the target automatically and explodes.

A number of military engineers videotaped the flight path of six Stinger missiles. They identified a number of ordered pairs to be able to trace the flight path on a Cartesian plane. Their results are in the tables below.

1 Plot each set of ordered pairs on the grid to estimate where each missile left the visible video screen. All of the missiles are launched from the origin (0, 0).

a

1	2	2	3	4	5	6	7	8	8.5
4	5	9	10	12	11	10	12	14	15

b

1	2	4	6	8	12	16	17	18	19
1	2	3	4	5	7	8	9	10	13

c

3	5	7	10	12	14	15	16	17.5	19
1	1.5	2	3	4	5	5	5.5	6.5	9

d

1	2	3	4	6.5	8.5	10.5	12.5	14	16
2	4	5	6	7.5	9	10.5	12.5	14	15

e

3	7	8.5	10	11	12	13	14	16	18
3	6	7	8	8.5	9	9.5	10	11	12

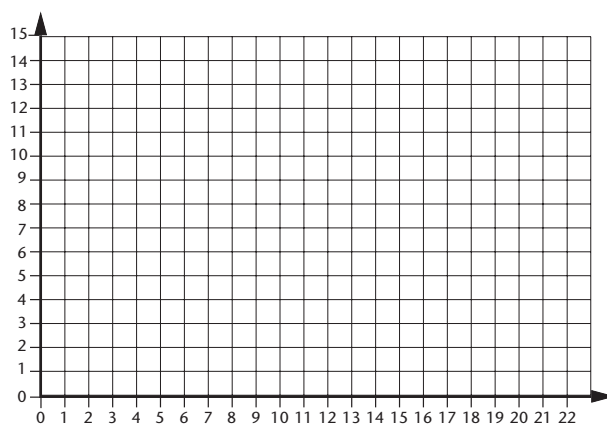
f

2	4	6	7.5	9	10.5	12	14	16	19
0.5	1	1	1.5	2	2.5	3	3.5	4	5

- 2 Which missile/s had the best flight path? Provide some reasons for your decision.
- 3 Which missile/s had a trajectory that had a partly vertical flight path?
- 4 Which missile/s had a trajectory that had a partly horizontal flight path?

The main rocket engine of a Stinger missile propels it to 2400 km/h. The Stinger missile can hit targets flying as high as 3500 m, and has a range of about 8 km.

- 5 Convert 3500 m to kilometres.
- 6 Convert 8 km to metres.
- 7 How long in seconds does it take the Stinger to reach a target located at a distance of:
 - a 3500 m?
 - b 8 km?
 - c 2.5 km?
 - d 4500 m?





The military engineers then videotaped the flight path of six ballistic missiles and identified their flight paths with the formulas shown. Each missile is released from the origin.

8 Complete the tables for each missile and plot the ordered pairs on the Cartesian plane. Use this to decide which missiles clear the mountain and are able to travel at least the minimum distance specified by the manufacturer. In the formulas, H is the height the rocket reaches in metres and D is the horizontal distance travelled, to the nearest kilometre. The mountain is situated 15 km from the origin and is 70 m high. Write your answers to the nearest unit.

a $H = 20D - D^2$, min. distance 15 km

D	1	2	3	4	5	10	11	16	18	19
H	19			64			99			

c $H = 12D - 0.75D^2$, min. distance 22 km

D	1	2	3	4	5	10	11	13	14	15
H			29					29		

e $H = 19.8D - 0.9D^2$, min. distance 20 km

D	1	2	3	6	9	10	11	17	18	21
H				86						

b $H = 12.5D - 0.5D^2$, min. distance 30 km

D	2	5	8	9	12	14	18	20	22	24
H		50					63			

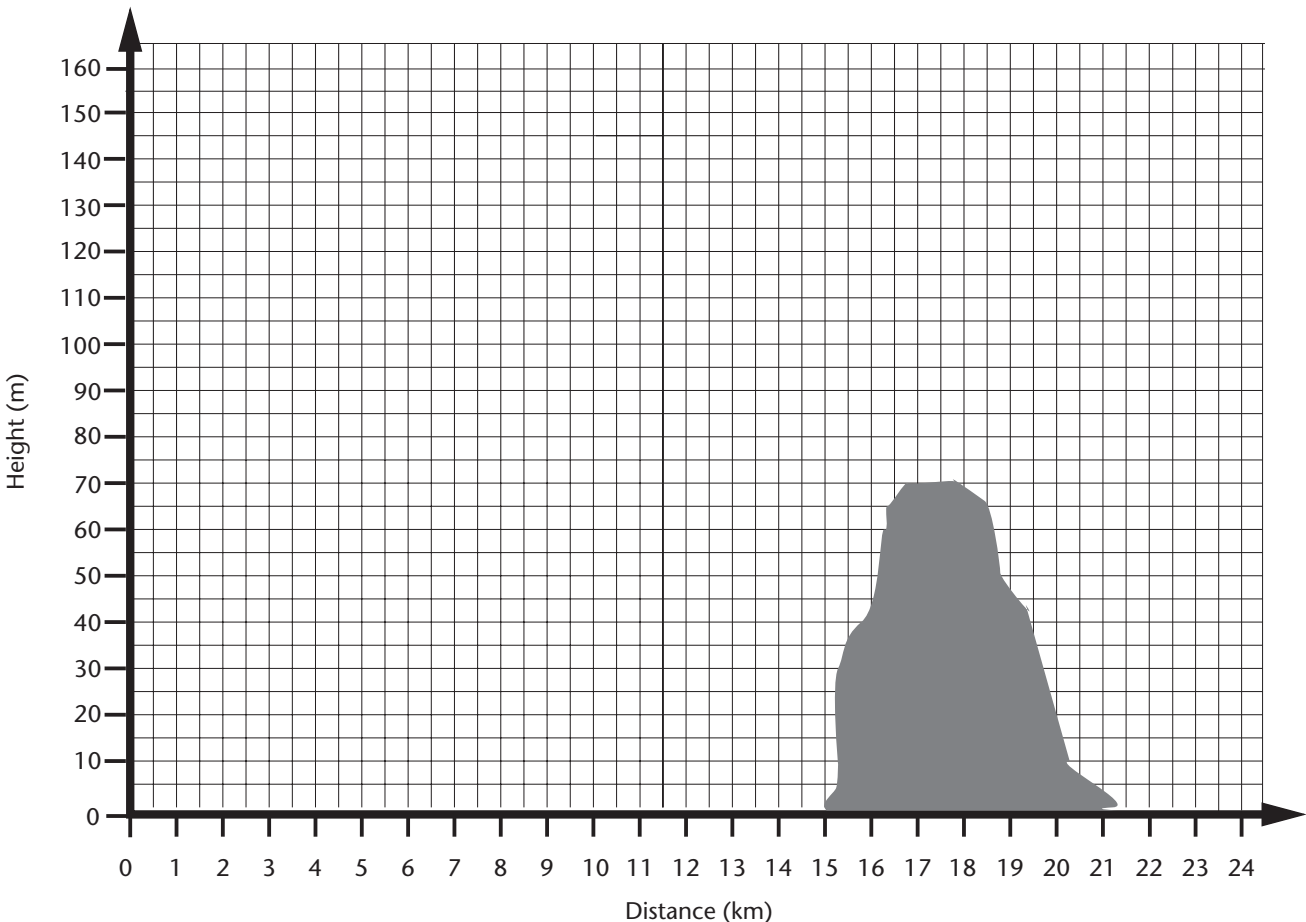
d $H = 14.4D - 0.8D^2$, min. distance 15 km

D	1	3	5	7	8	11	12	13	15	16
H			52							

f $H = 24.75D - 0.99D^2$, min. distance 23 km

D	2	4	8	9	11	15	17	18	21	24
H		83				149				

- 9 Which missiles could achieve the minimum distance specified?
- 10 Write down the maximum height each missile reached and whether it cleared the mountain.





Grid references are used by many people including the armed forces to locate places of interest particularly on maps. Grids lines are placed on maps and labelled with sequential numbers or letters to provide a reference to features on the map. The numbers or letters are known as grid references. For example, Geranium is located in D3.

1 Name the city at each grid reference.

- a** E 2 **b** F 3
c F 4 **d** D 3

2 Find the grid reference of:

- a** Lithgow
b Costa
c Mandarin
d Lipton.

3 Name the country at each grid reference.

- a** D 2 **b** D 4
c E 2 **d** F 3

4 Find the grid reference of:

- a** Bonzai **b** Ablett
c Muscato **d** Jeronto.

5 Use the scale to calculate approximately the length and width of the grid at E3.

The squares on a grid do not accurately locate the position of a city. Lines of latitude (running in an east–west direction, e.g. 5°S) and longitude (running in a north–south direction, e.g. 10°N) are used to specify more accurately the location of a destination on the map. The position for Geranium could be specified as 49°N 2°E or even more accurately as 48.52°N, 2.20°E.

6 Name the place of interest at:

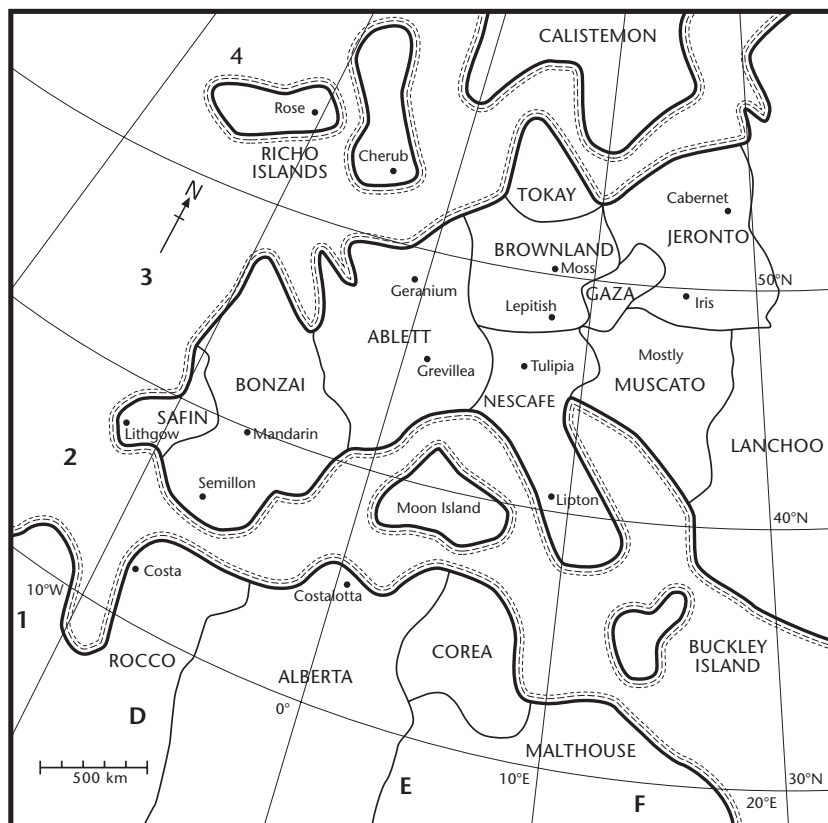
- a** 41°N, 40°W **b** 53°N, 8°W **c** 51°N, 12°E **d** 50°N, 19°E.

7 Write down the location of:

- a** Cherub **b** Lipton **c** Lepitsh **d** Costalotta.

8 A ballistic missile has a range of 1500 km and is launched from the outskirts of Cherub. Which of these cities can it reach?

- a** Grevillea **b** Semillon **c** Tulipia **d** Iris





- 9** Military leaders are planning to defend the countries on the map from attack. They have to decide between Moss and Mandarin as the better site to locate their missile launcher. Name the cities they can defend from each launch site and decide which city would be the better location. The range of the missiles is 1500 km.

Moss	
Mandarin	

The better location is_____.

Military tacticians often need to specify the distance and direction that their personnel must take to reach a certain destination. Directions can be specified using True Compass Bearings which are measured clockwise from north. For instance, the true bearing from Cherub to Geranium is approximately 150°T .

- 10** Military personnel are to travel from Grevillea in the country of Ablett to each of these cities. Specify the true bearing they must travel.

- | | |
|-------------------|---------------------|
| a Cherub | b Costalotta |
| c Iris | d Mandarin |
| e Cabernet | f Moss |
| g Costa | h Lepitsh |

- 11** Military personnel are to travel from Lithgow in the country of Safin to each of these cities. Specify the distance and direction they must travel.

- | | |
|------------------|-------------------|
| a Rose | b Lipton |
| c Cherub | d Cabernet |
| e Tulipia | f Mostly |
| g Lepitsh | h Moss |

- 12** The military personnel in Q11 are travelling in a stealth plane, which can fly at a speed of 880 km/h. How long does it take each person to reach their destination?

- 13** A stealth plane takes off from Semillon at a bearing of 38°T and travels 1500 km. Did the plane reach a destination? If not, by how much did it miss?

- 14** Another stealth plane takes off from Costalotta on a bearing of 80°T and travels a distance of 1400 km. Where does it land?

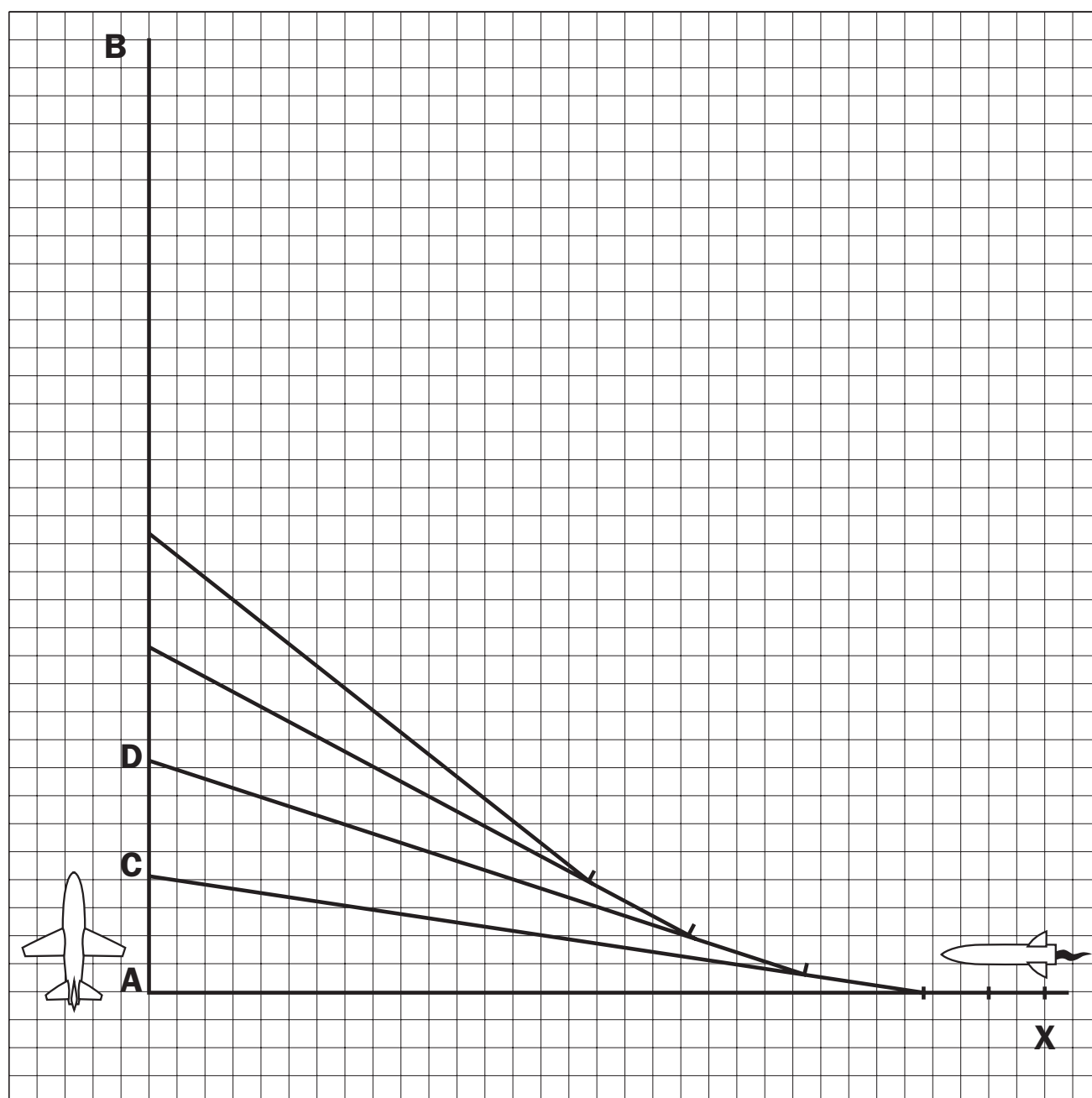
- 15** Select a secret site for a stealth plane to take off from an aircraft carrier based at a secret location at sea. For instance, $35.00^{\circ}\text{N } 18.00^{\circ}\text{E}$. Specify the range and distance to several cities, and ask a friend to check your answers.





Pierre Bouger in 1693 was the first mathematician to seriously study curves of pursuit. Earlier, Leonardo Da Vinci had also dabbled in this area.

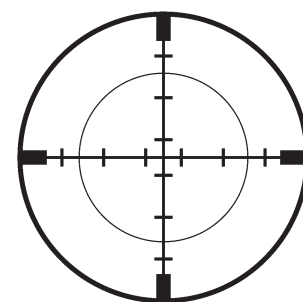
Imagine a jet fighter plane travelling in a straight line from point A to point B and a heat-seeking missile being launched from point X. Both the missile and the jet fly at the same speed, which means that both will fly the same distance in any given time. After 30 seconds the missile changes course to hit the jet at C, after another 30 seconds the missile again changes course to hit the jet at D and so on.



- 1 Use a ruler and pencil to complete the pursuit curve. Colour every second triangle to make a design.
- 2 Is it possible for the missile to catch the jet fighter? Explain your answer. (Hint: Extend your graph past point B above).

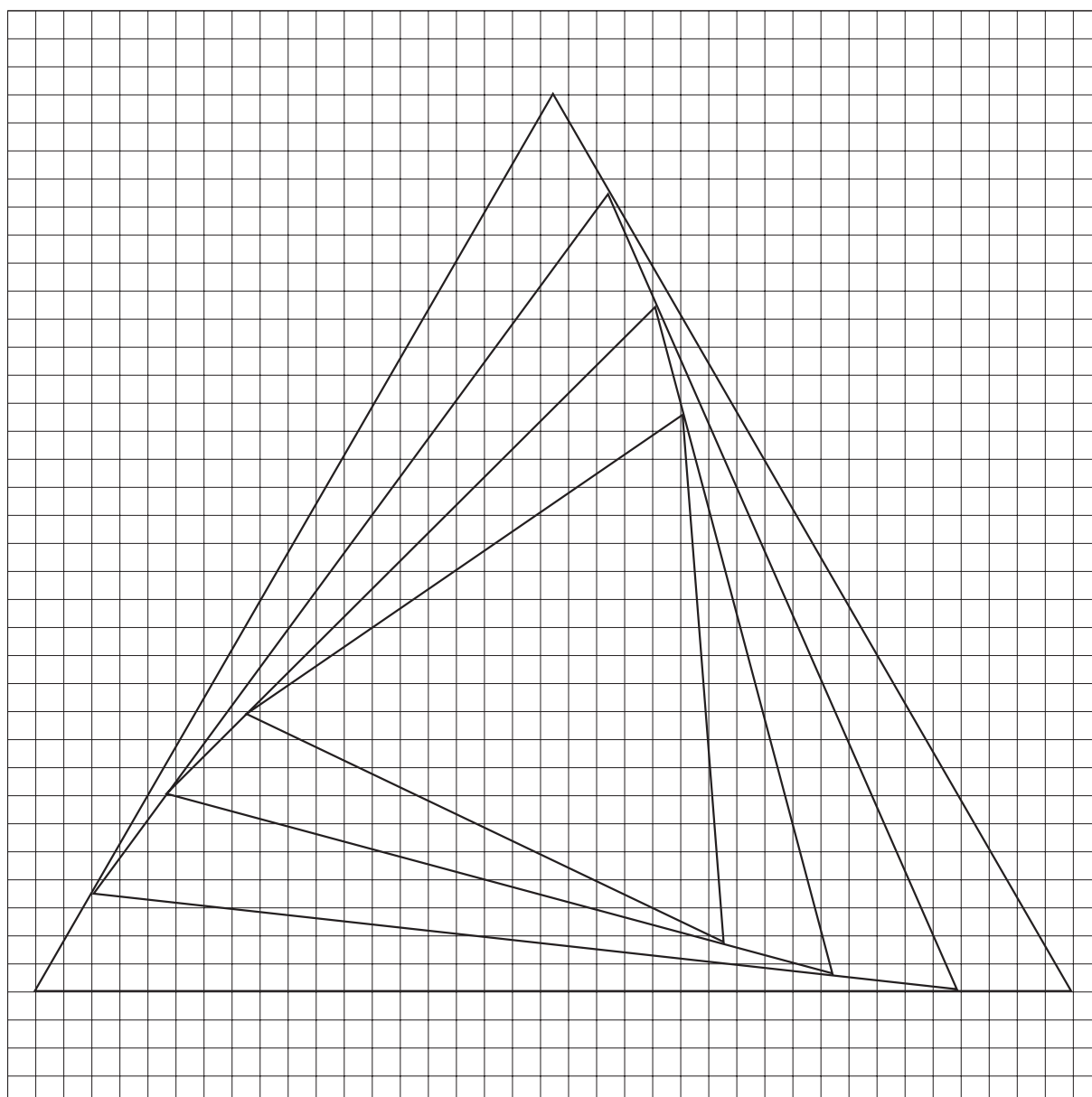


Obtain some graph paper from your teacher, preferably 1 cm square grid. Use smaller intervals than the example on the previous page.



- 3 Draw pursuit curves to investigate what happens when the:
 - a missile flies at double the speed of the jet
 - b jet flies at double the speed of the missile.
- 4 In the example the initial flight paths are at right angles. Vary the angle of the initial flight paths and draw the pursuit curves.
- 5 An equilateral triangle is drawn below. A soldier is located at each of the vertices, and each can run at the same maximum speed. Each soldier pursues the soldier to his right. Draw as many triangles as you possibly can. Highlight the pursuit curves and colour a pattern.
- 6 Repeat this if the original shape is a:

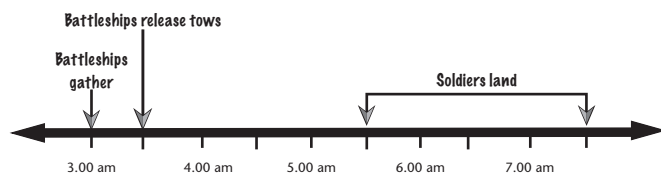
a square	b pentagon	c hexagon	d scalene triangle
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Selected answers

1 Gallipoli and the Anzacs

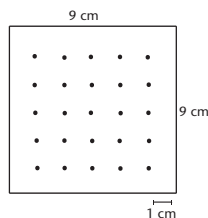
- 2 a ≈ 33 km b ≈ 6 km 3 a $40^{\circ}17'N$ $26^{\circ}14'E$
 b $40^{\circ}13'N$ $26^{\circ}18'E$ c $40^{\circ}25'N$ $26^{\circ}42'E$
 d $40^{\circ}8'N$ $26^{\circ}26'E$ e $40^{\circ}4'N$ $26^{\circ}13'E$
 6 a 15 000 b 45 000 c 25 000 d 90 000
 7 a 0330 b 0300 c 0530 d 0730 e 2030
 8



- 9 144 10 100 m² 11 a 73 000 b 27 000 c 28 000
 d 8000 e 4800 f 140 g 252 000 h 131 450, 261 490, 392 940 12 1 651 over 238 days

2 Hard tack and the Anzacs

- 1 15 cups white flour, 3 cups wholemeal flour, 5 tablespoons sugar, 3 tablespoons milk powder, good pinch salt, 1 cup water 2 a 21 g b 42 g c 56 g d 91 g
 3 a 250 g b 375 g c 562.50 g d 775.00 g 4 a 3.2 cups b 6.4 cups c 3.6 cups d 4.96 cups
 5 a 8 g b 16 g c 48 g d 28 g 9
 6 a 85.05 g cheese, 17.72 g tea, 113.4 g jam, 14.18 g salt, 0.79 g pepper, four hard tack biscuits 85.05 g each, 396.9 g bully beef 7 a 11.71% b 1.83%
 c 35.14% d 40.99% 8 6.8 t hard tack, 2.3 t jam

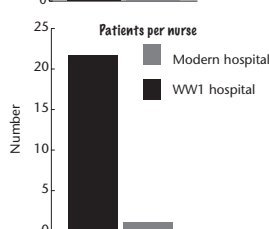
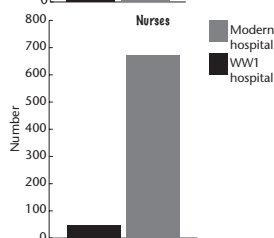
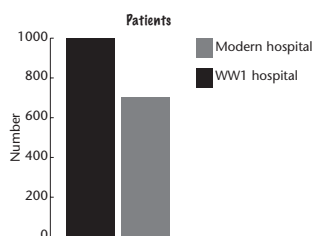
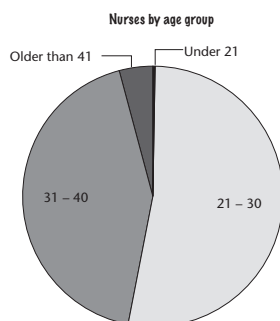


3 The Western Front

- 2 620 km 620 000 m 3 G = 1 900 000, F = 2 150 000, B = 600 000, Bel = 100 000, A = 50 000 4 4 800 000
 5 G = 39.58%, F = 44.79%, B = 1.25%, Bel = 2.08%, A = 1.04% 6 a ≈ 110 km b ≈ 90 km 7 a ≈ 230 km
 b ≈ 280 km c ≈ 270 km d ≈ 260 km 9 a B4 b D3
 c A6 d E2

4 The nurses

- 1 2560 2 a 83.59% b 16.41% 4
 c 15.23% d 0.98% 3 a 1000, 700 b 1000, 700 c 46, 670
 d 21.74, 1.04 5 a 7/2227, 1°
 b 1180/2227, 191° c 950/2227, 154° d 90/2227, 15° e 2227



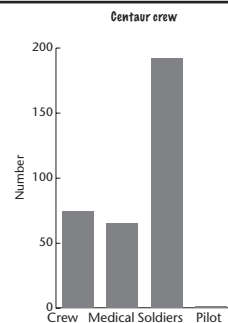
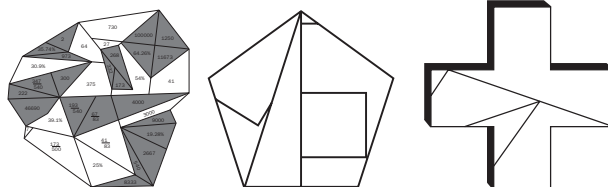
5 Red Cross

- 1 a 1 200, 25 000, 6/125, 4.8% b 8 500, 20 000, 17/40, 42.5% c 1 200, 10 000, 3/25, 12% d 2 400, 40 000, 3/50, 6% 2 a 7 b 2 3 a 80 000 b 6 667 4 a 7 200 b 138

- 5 46 690 km 6 a 9 338 km 17
 b 778 km 7 a 167 b 2 000
 8 a 4 000 kg b 300 kg
 9 More than 1250 10 40
 11 a 347/540 b 193/540
 12 a 35.74% b 64.26% 13 332
 14 268 15 67/83 16 19.28%

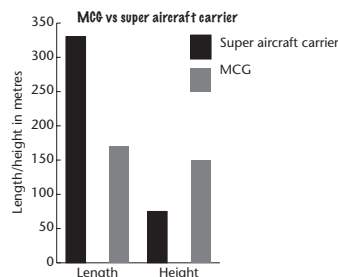
19 Blood donations, helping people in tragic situations, support elderly, provide breakfast for hungry children etc.

18



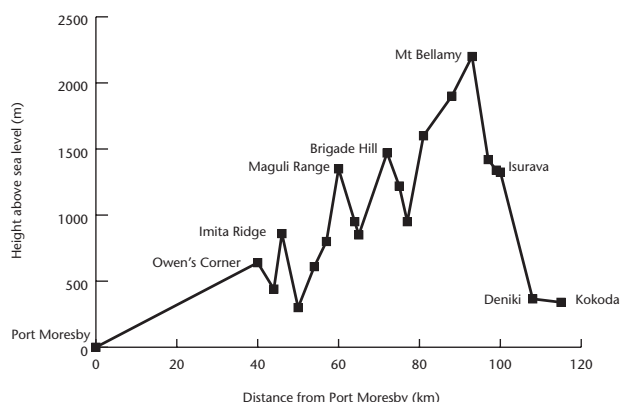
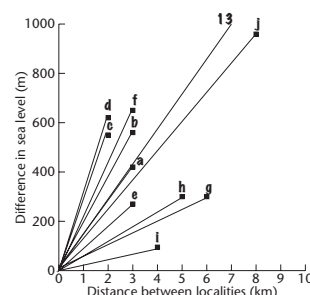
6 The Battle of the Coral Sea

- 2 a $10^{\circ}N$ $125^{\circ}E$ b $21^{\circ}N$ $158^{\circ}W$ c $35^{\circ}N$ $135^{\circ}E$ d $5^{\circ}N$ $175^{\circ}E$
 e $15^{\circ}S$ $155^{\circ}E$ f $5^{\circ}S$ $155^{\circ}E$ 3 a ≈ 4 900 km, 75.38 h b ≈ 5 800 km, 89.23 h c ≈ 7 200 km, 110.77 h d ≈ 1 000 km, 15.38 h e ≈ 6 200 km, 95.38 h 4 99 5 52.11%
 6 56.57% 7 23.75
 9 107 10 54.08% 11
 12 a 4.80 b 24 c 5
 13 a 2 min 5 s b 2 min 30 s
 c 4 min 10 s d 1 min 15 s
 14 a 61 000 t b 310 t
 c 160 kg d 300 000 kg each e 125 million L
 f 6000 g 800
 h 180 000 L i \$45 billion



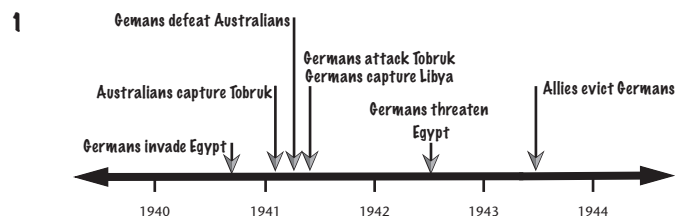
7 The Kokoda Trail

- 1 a $39^{\circ}N$ $139^{\circ}E$ b $9^{\circ}S$ $148^{\circ}E$ c $8^{\circ}S$ $115^{\circ}E$ d $12^{\circ}S$ $132^{\circ}E$
 e $16^{\circ}S$ $145^{\circ}E$ f $15^{\circ}N$ $120^{\circ}E$ 2 a Australia b Philippines
 c Indonesia d Japan e Vietnam f Taiwan 3 a ≈ 7 200 km
 b ≈ 2 600 km c ≈ 2 000 km d ≈ 1 200 km e ≈ 4 400 km
 f ≈ 4 100 km 4 a 172° b 217° c 220° d 170°
 5 69 years 6 828 months 10
 9 a 3 km, 420 m b 3 km, 560 m c 2 km, 550 m
 d 2 km, 620 m e 3 km, 270 m f 3 km, 650 m
 g 6 km, 300 m
 h 5 km, 300 m
 i 4 km, 95 m
 j 8 km, 959 m
 11 Menari to Brigade Hill
 12 Iora creek to Isurava
 14



Selected answers (cont'd)

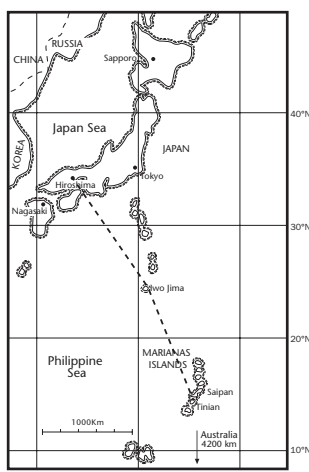
8 The Rats of Tobruk



- 3 858 500 4 a 440/1717 b 37/1717 c 1240/1717
 5 a 25.63% b 2.15% c 72.22% 6 13 days 7 38 500
 8 a 1038.46 b 1923.08 9 a 6.75% b 22.73% 10 a 1/3
 b 2/3 11 93, 31 12 Ammunition, food, water, vehicles,
 soldiers 13 151 51.52 t supplies. 1212.12 vehicles
 14 20% supplies, 14.71% vehicles 15 26 000 16 a 7/13
 b 6/13 17 53.85% 18 8 months 10 days 20 a 1230 h
 b 0230 h c 1715 h d 0130 h 21 a 6.00 pm b 2.30 am
 c 11.00 pm d 12.30 pm

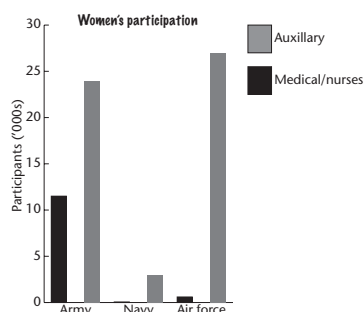
9 Little Boy and Fat Man

- 2 a 15°N145°E b 32°N130°E 4
 c 44°N142°E 3 a Iwo Jima
 b Saipan c China 5 5600 km
 6 46.667 km/h
 7 a 2.44 km b 9.855 km
 8 7.42 km 10 0815 h
 11 00.6 t 12 0.95 min
 13 162.37 m/s
 14 a 828 g b 59 172 g
 15 8.04 km² 16 8 040 000 m²
 17 3742.41 m 18 150 000
 19 53.33% 20 a 48 000
 b 24 000 c 8 000
 21 11.01.43 am 22 1101 h
 23 22.22% 24 6 400 g 25 144 h



10 The changing roles of women in war

- 1 Infection, illness 2 2% 5
 3 926 500 4 a 35 500
 b 3 060 c 27 600
 d 66 160 6 66 160
 7 a 53.66% b 4.63%
 c 41.72% 8 a 23/71
 b 1/51 c 1/46
 9 a 67.61% b 98.04%
 c 97.83% 10 Domestic
 workers, Tradespeople, Skilled
 professional, Shop assistants,
 Farm workers 11 ≈6.96% 12 Domestic workers,
 Tradespeople, Skilled professional, Shop assistants,
 Administration 13 ≈5.96% 14 ≈6.1%



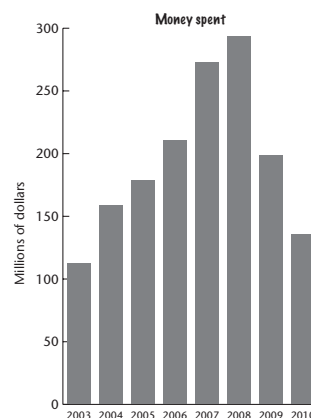
- 15 In some categories there has been a large increase in
 the percentage participation rate for women
 16 Men returned from the war and took these jobs
 17 a ≈74% b ≈38% c There has been a large increase in
 the participation rate for older women d There has been a
 large increase especially in the 20 to 35-year-old group.

11 The Vietnam War

- 4 a 1 563 400 b 285 427 c 513 300 d 1 427 466
 5 a Y,Y b Y,N c Y,N d N,N e Y,Y f N,N
 6 a All caves have an even number of tunnels leading
 from them, start and finish at any cave b If two caves have
 an odd number of tunnels leading from them then start
 and finish at the odds 7 a start at B finish at H b start at B
 finish at D c Start and finish at any point

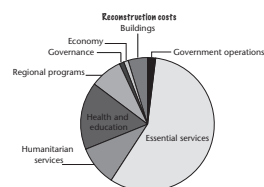
12 The war in Iraq

- 1 a 0.08% b 0.08% 3
 c 0.01% d 1.74% 2 Iraq
 contributed the most soldiers,
 Australia the least
 5 \$1.556 billion
 6 \$1 556 000 000 7 \$71.05
 8 \$8.88 9 a 0.44%
 b 0.27% c 7 d 5
 e 0.11% f 0.69% g 30
 h 3.75% i 0.06% j 3 k 4
 l 0.38% m 2 n 92.27%
 o 0.23% p 0.19%
 12 a ≈475 km b ≈550 km
 c ≈100 km d ≈310 km
 13 a ≈141° b ≈352° c 20° d 56°



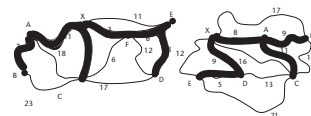
13 Rebuilding Iraq

- 1 a 7° b 206°
 c 34° d 59°
 e 28° f 4° g 4°
 h 17°
 2 \$38.871 billion
 4 XDEFGHABCX
 5 EFGXAHBCDE 6 Yes
 7 a 41 km b 41 km
 8 a 51 km ADCHGBFE
 b 51 km AHGDCBFE
 9 a 40 km b 41 km
 10 a \$2 296 000 b \$2 296 000



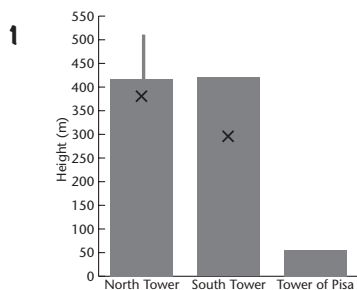
3

	1	2	3	4	5	6
a		X			X	X
b					X	



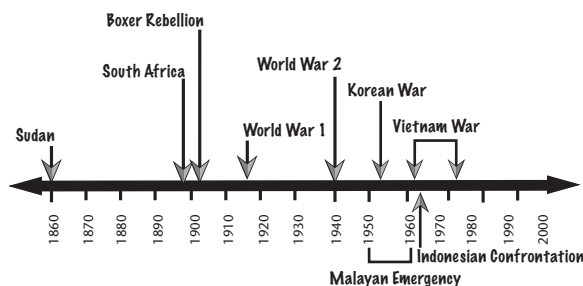
14 The war on terror begins

- 3 196 000 000 kg
 4 510 000 000 kg
 5 777 600 t 6 87 200
 windows 7 a 19 300 000 m
 b 48 250 times 8 a 8.46 am
 b 9.03 am c 9.59 am d 10.28
 am e 5.20 pm f 5.21 pm
 9 a 102 min b 6120 s c 1.7 h
 10 a 56 min b 3360 s
 c 0.93 h 11 515 min
 12 \$43.32 13 \$35.52



15 Australian war casualties

- 1 a 1.17% b 1156, 7.23% c 6, 1.07% d 155 000,
 52.53% e 132 253, 12.98% f 339, 10.45% g 56, 08%
 h 15, 0.43% i 2 398, 5.79%
 3



- 4 Deaths 1.5%, 0.50%; captures 0.10%, 0.44%;
 injuries 3.88%, 0.96% 5 a 43500 b 27500 c 71000
 6 a 24% b 29% c 18% d 20% 7 a 3:1 b 2:1 8 9% 9
 Answers will vary.

Selected answers (conf'd)

16 Knots and naughty miles

1 360° 2 21 600' 3 21 600 NM 4 39 488.64 km
 5 1.83 km 6 a ≈ 3 420 NM b ≈ 2 340 NM c ≈ 2 900 NM
 d ≈ 4 740 NM e ≈ 2 280 NM 7 a 10 kn b 7.5 kn 8 a 1 h
 b 28 h 34 min c 34 h 17 min 9 a 11 h 50 min b 5 h 50 min
 c 41 h 54 min 10 a 33.33 kn, 61 km/h b 49.18 kn,
 90 km/h c 32.29 kn, 5908 km/h 11 a 4 920 NM, 6d 20h
 b 3 360 NM, 4d 16h c 5 580 NM, 7d 18h d 3 600 NM,
 5 d e 2 700 NM, 3 d 18 h 12 a 0, Pearl Harbour
 b 3rd Jan, 1 980 NM, 2d 9h, at sea c 4 January, 900 NM,
 1d 2h, Samoa d 6 January, 1 680 NM, 2d, Nauru
 e 8 January, 1 620 NM, 1d 22h, Micronesia f 12
 January, 3 000 NM, 3d 14h, Midway 13 10d 23h
 14 15.25 m

17 The tank

2 a 2 h b 3 h
 c 5 h d 8 h 45
 min 3 a 105
 km b 175 km
 c 91 km d 87.5
 km 4 a 300 cm
 b 200 cm c 245
 cm d 354 cm
 5 a T b F c T d F

6 The river may be too deep and they might drown

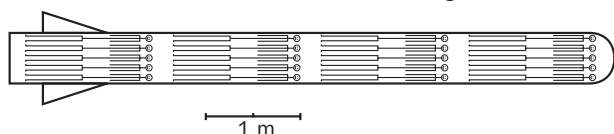
7 Surprise attack on the enemy

8 a 155.56 L/100 km

b 280 L/100 km c On the road the fuel consumption of a tank is ≈ 15 times that of a car and cross country it is ≈ 28 times.

18 Missile mayhem

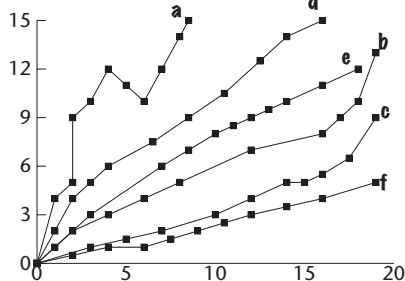
1 a 261 cm b 234 cm c 250 cm
 d 503 cm 2 a 2612 mm b 2340 mm c 2300 mm
 d 2130 mm 3 a 800 000 m b 1 200 000 m
 c 1 500 000 m d 1 340 600 m 4 \$750 000 5 a \$2.25
 million b \$4.5 million c \$9 million d \$17.25 million
 6 a 0.9 t b 2.7 t c 7.2 t d 15.3 t 7 a 2 b 4 c 8 d 10
 8 a 2 b 1 c 6 d 10 9 a ≈ 1.5 m b ≈ 50 kg 10 a 4 b 30
 11



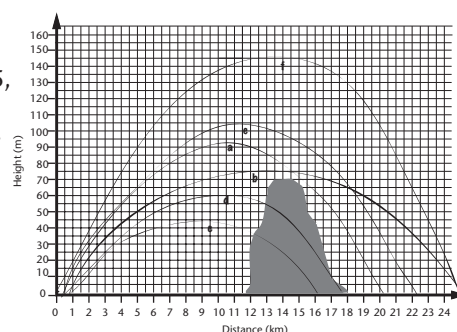
12 a 23 000 g b 45 000 g c 64 600 g d 450 100 g
 13 a 43 kg b 12 kg c 650.1 kg d 400.6 kg 14 9.23
 15 a 40 000 mL b 600 000 mL c 500 600 mL d 650
 200 mL 16 a 6 L b 320 L c 610.8 L d 563.2 L 17 a 1
 760 km b 2 640 km c 3080 km d 2 200 km 18 a 600
 km/h b 866.67 km/h c 880 km/h d 1 184.21 km/h

19 Ballistic and Stinger missile

1
 2 Missiles e and f had the most direct and straightest flight paths
 3 Missile a
 4 Missile c
 5 3.5 km
 6 8000 m
 7 a 5.25 s
 b 12 s
 c 3.75 s d 6.75 s 8 a 36, 51, 75, 100, 64, 36, 19 b 23, 68,
 72, 78, 77, 50, 33, 12 c 11, 21, 36, 41, 45, 41, 21, 11



d 14, 36, 62, 64,
 62, 58, 52, 36, 26
 e 19, 36, 51, 105,
 108, 109, 77, 65,
 19 f 46, 135, 143,
 152, 135, 125, 83,
 24 9 a, d, e, f
 10 a ≈ 100 m, Y
 b ≈ 78 m, Y
 c ≈ 46 m, N
 d ≈ 65 m, N
 e ≈ 108 m, Y
 f ≈ 155 m, Y

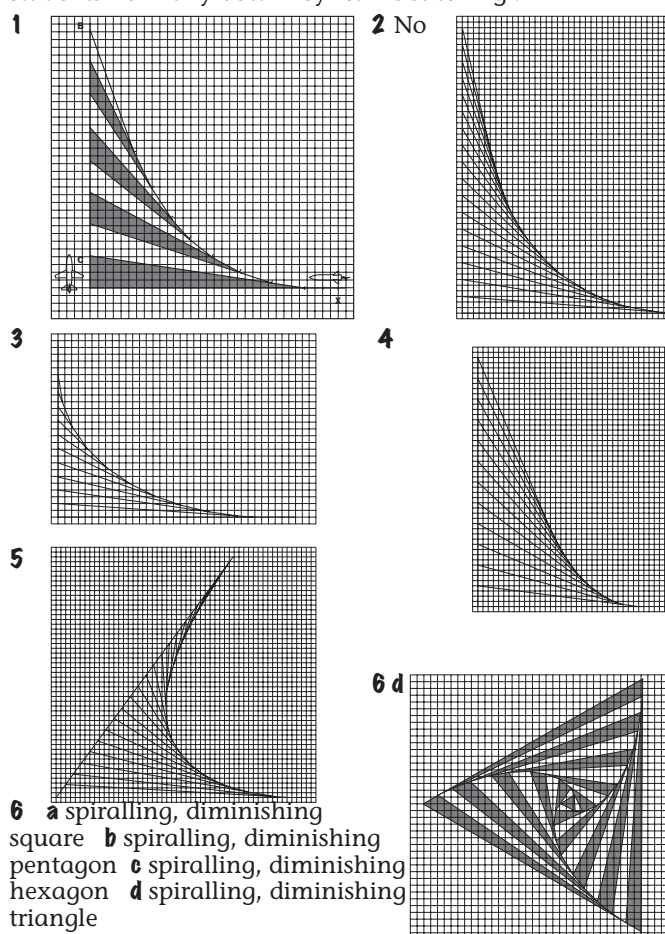


20 Mapping madness

1 a Costalotta b Mostly c Iris or Moss d Mandarin
 2 a C2 b D2 c D3 d F3 3 a Rocco b Richo Islands
 c Alberta or Corea d Nescafe, Muscato or Gaza
 4 a D3 b E3 c F3 d F4 5 900 × 1100 km
 6 a Mandarin b Rose c Moss d Iris 7 a 54°N1°W
 b 42°N12°E c 48°N11°E d 36°N3°E 8 a Y b N c Y
 d Y 9 Mandarin: Tulipia, Grevillea, Lepitsch, Geranium,
 Lithgow, Semillon, Costa, Costalotta Moss: Cherub, Rose,
 Caberent, Iris, Mostly, Lipton, Tulip, Lepitsch, Grevilea,
 Geranium The better location is Mandarin as more cities
 can be protected 10 a ≈ 330°T b ≈ 190°T c ≈ 58°T d ≈ 236°T
 e ≈ 48°T f ≈ 40°T g ≈ 221°T h ≈ 56°T 11 a ≈ 1700 km, 4°T
 b ≈ 2000 km, 73°T c ≈ 1700 km, 25°T d ≈ 2800 km, 45°T
 e ≈ 1800 km, 56°T f ≈ 2300 km, 55°T g ≈ 2000 km, 51°T
 h ≈ 2100 km, 45°T 12 a 1.93 h b 2.27 h c 1.93 h
 d 3.18 h e 2.05 h f 2.61 h g 2.27 h h 2.39 h 13 No,
 the missile travelled too far, assuming the destination is
 Grevillea 14 Buckley Island.

21 Pursuit curves

Please note that the curves produced here are not those students normally obtain by 'curve stitching'.



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